

Arbitrage-Free Deep Option Pricing via Convex Composition: Architectural Guarantees and Universal Approximation

[Author Name]¹[0000-0000-0000-0000]

[Department, Institution, City, Country]
author@institution.edu

Abstract. We introduce ARBNET, a neural call-pricing architecture that is free of static arbitrage *by construction* rather than by penalisation. The construction parameterises the European call price as the sum of a forward-intrinsic baseline and a non-negative correction, where the correction is expressed as a finite mixture of products of (i) input-convex networks in the strike argument and (ii) monotone networks in time-to-maturity, modulated by a smooth boundary factor that vanishes at expiry. The composition machinery is elementary — convexity is preserved under non-negative weighted sums and under composition with convex non-decreasing scalar functions, and the product of two non-negative non-decreasing functions is non-decreasing — but it suffices to deliver four pointwise no-arbitrage conditions simultaneously: absence of butterfly arbitrage, absence of calendar arbitrage, the expiry boundary, and the forward-intrinsic lower bound. We prove that the resulting surface satisfies these conditions for every choice of network weights, with no tolerance-based residual. We further prove a universal-approximation theorem on the natural hypothesis class — the cone of continuous, strike-convex, time-monotone, non-negative surfaces vanishing at $T = 0$ on a compact strike-maturity box — with an explicit approximation rate governed by the bivariate modulus of continuity of the target. We honestly characterise the expressivity boundary of the construction: surfaces whose forward time-value is non-convex in strike, including the entire Black–Scholes family at any positive volatility, lie outside the hypothesis class and cannot be represented exactly. Empirically, on a population of 30 rough-Bergomi surfaces at Nifty-50 spot level, the architecture achieves identically zero butterfly and calendar violation rates on every (surface, seed) trial, against a soft-penalty Akerer-style baseline whose mean calendar-violation rate is 0.81% with maximum 1.83%. The compliance guarantee comes at a price-fitting cost of approximately 56.7 INR in mean RMSE relative to the soft-penalty baseline. We argue, with figures and statistics, that this fit-compliance frontier is structural rather than artifactual: it reflects the hypothesis-class restriction made by the convexity constraint, and is the unavoidable cost of moving compliance from a tolerance-bound regulariser to an architectural invariant. The architecture is therefore appropriate whenever downstream consumers (hedging desks, value-at-risk pipelines, exotic-product calibration) re-

quire certified no-arbitrage at every grid evaluation, regardless of the training trajectory.

Keywords: No-arbitrage · Neural option pricing · Input-convex networks · Monotone networks · Universal approximation · Deep hedging · Rough volatility · Risk-neutral density.

JEL classification: C45, G13.

MSC 2020: 91G20, 91G60, 68T07.

1 Introduction

1.1 Context and motivation

A European call surface, viewed as a function $(K, T) \mapsto C(K, T)$ at a fixed valuation date, encodes the market’s joint expectation of strike-payoff and maturity-payoff over the underlying. Modern derivative desks consume this surface in three distinct downstream tasks: hedging (delta and gamma extraction via the first two strike derivatives of the surface and the spot derivative of the price); marking the book and computing capital (the surface determines the risk-neutral density via the Breeden–Litzenberger relation [16], which feeds value-at-risk pipelines and stress tests); and calibrating exotic-product models (variance swaps, barrier options, and cliquets all require a consistent surface to extract a unique implied density). Each of these consumers depends on the surface being *free of static arbitrage*: no portfolio of calls at the same valuation date should yield a non-negative payoff at some future date with strictly negative cost today.

Static arbitrage on a call surface manifests as the violation of any of three pointwise structural conditions [30,78,34]. Convexity in strike at every maturity is implied by the existence of a butterfly portfolio with non-negative payoff but zero cost; its violation has been documented as the canonical no-arbitrage failure mode on stressed surfaces [65,39]. Monotonicity in maturity of the carry-adjusted price ($e^{rT}C(K, T)$ non-decreasing in T) is implied by the calendar-spread argument; its violation creates arbitrage between options of different maturities at the same strike. Finally, the forward-intrinsic lower bound $C(K, T) \geq e^{-rT}(F_T - K)^+$, with $F_T = Se^{(r-q)T}$, comes from the static replication of the forward via long-call–short-put parity [73]. A fourth, often implicit, condition is the expiry boundary $C(K, 0) = (S - K)^+$.

Classical financial theory provides parametric models that are arbitrage-free by construction. The Black–Scholes model [14] is arbitrage-free at any single volatility but cannot fit empirical surfaces, which exhibit pronounced smiles and skews [27,77]. The local-volatility model of Dupire [36] fits perfectly by construction but has been shown to mismatch dynamic properties of the surface [28,49]. Stochastic-volatility models (Heston [54], SABR [49], and their many extensions [5]) are dynamically arbitrage-free and have analytically tractable transforms [71,20,37], but their fit to empirical surfaces requires either many free parameters [6] or model extensions such as jumps [19,29]. Rough-volatility models [2,9,10,12,38] achieve realistic ATM volatility behaviour with few parameters

but have non-trivial calibration costs [58]. At the surface level, the SVI [42] and SSVI [43] parameterisations of total implied variance are arbitrage-free under explicit parameter constraints [44,45]; they are widely deployed in practice but have a finite-dimensional capacity that constrains the fit on stressed wings.

1.2 The static-arbitrage failure mode of neural pricers

The advent of deep learning has produced an array of neural pricers [1,23,59,79,80,61,63,83,84,72,40], which trade the parametric tractability of classical models for the capacity of multi-layer architectures. These pricers achieve excellent in-sample fit and, with enough training, decent out-of-sample generalisation. However, almost all neural pricers enforce no-arbitrage *softly*: an additional term proportional to the negative part of the local arbitrage violation is added to the training loss, typically as $\lambda_{\text{bf}} \mathbb{E}[(-\partial_K^2 C_\theta)_+^2]$ for the butterfly condition and $\lambda_{\text{cal}} \mathbb{E}[(\partial_T(e^{rT} C_\theta))_-^2]$ for the calendar condition, with the expectations approximated by Monte Carlo on a finite training grid. The soft penalty paradigm has a number of well-understood limits.

First, the penalty is a regulariser, not a constraint. At any finite penalty weight λ , the optimum trades off the pricing-error term against the penalty term, and as long as the pricing-error gradient pulls the surface into the arbitrage region anywhere, some violation will persist. In our experiments, with the standard penalty weight $\lambda = 1$ and training budgets in the hundreds of epochs, the residual calendar-violation rate of the soft-penalty baseline averages 0.81% of grid points with a maximum of 1.83%.

Second, the penalty is evaluated on the training grid, not over the continuous domain. The neural pricer is queried at many points off-grid by downstream consumers — e.g., at the strikes of exotic-product instruments, at fine grids for risk-neutral density extraction, or at perturbations of the spot for delta hedging — and at those off-grid points, no penalty was applied during training. A surface that is compliant on the training grid can therefore violate the conditions arbitrarily far off-grid. This is the noisy-empirical-grid analogue of the more general phenomenon studied by Aït-Sahalia and Duarte [7] in the context of shape-constrained nonparametric pricing.

Third, residual violations propagate. A non-convexity in K at one grid point implies a negative second strike derivative, which in turn implies a negative implied density at that point. The negative density propagates into any computation that uses the surface — expected payoffs of exotic products, value-at-risk computations, model-free volatility indices — with potentially significant aggregate effects. Hedging with a delta extracted from a non-convex surface introduces additional path-dependent variance because the delta is no longer monotone in strike.

The natural alternative is to make the architecture itself arbitrage-free. This idea has historical precedent. SVI [42] and SSVI [43] are structurally arbitrage-free under parameter constraints in the implied-variance domain. In the call-price domain, Kahalé [65] and Fessler [39] construct arbitrage-free interpolators of finite quote sets via spline-based monotone-convex schemes; the constructions

are exact but not learnable from data at the level of a neural model. Density-network approaches [59] parameterise the risk-neutral density ρ_T directly; this trivially enforces butterfly but does not give a closed-form calendar enforcement. Neural stochastic differential equation (neural-SDE) market models [26,25,31,41] build arbitrage-free dynamics on the underlying process; static no-arbitrage is then a corollary. This last line is the most theoretically satisfying but is also the most computationally costly and the least transparent to inspect at the level of a single surface.

1.3 Contributions

We make four contributions, each pinned to an explicit claim and supported by theory or experiment.

C1. A neural architecture with structural arbitrage-freeness. We parameterise the call surface as

$$C_\theta(K, T) = e^{-rT} [(F_T - K)^+ + \Delta_\theta(K, T)], \quad F_T = S e^{(r-q)T}, \quad (1)$$

with a non-negative correction Δ_θ given by

$$\Delta_\theta(K, T) = S(1 - e^{-\alpha_\theta T}) \sum_{j=1}^J s_{j,\theta} \text{softplus}(\phi_j^{\text{cvx}}(K/S)) \text{softplus}(\phi_j^{\text{mon}}(T)), \quad (2)$$

where each ϕ_j^{cvx} is an input-convex network [3,18,24] in normalised strike, each ϕ_j^{mon} is a monotone network [33,82,86,81,48] in time-to-maturity, and the scalar gates satisfy $s_{j,\theta}, \alpha_\theta > 0$. Theorem 1 establishes that this construction satisfies all four conditions of Definition 1 for every θ , with no tolerance bound. The construction is in the call-price domain rather than the implied-vol domain: as we discuss in Section 4.5, the implied-vol domain would require enforcing the non-linear Durrleman positivity condition, which is not preserved by the simple compositions used here.

C2. A universal-approximation theorem with explicit rate. Theorem 2 states that the hypothesis class $\mathcal{C}_{\text{cvx}, \nearrow}$ of continuous, strike-convex, time-non-decreasing surfaces vanishing at $T = 0$ on the compact strike-maturity box $\mathcal{B} = [K_0, K_1] \times [0, T_{\max}]$ is uniformly approximable by the parameterisation Equation (2). The number of mixture components scales as $J = \mathcal{O}(\omega_g^{-1}(\varepsilon/3)^{-2})$, where ω_g is the bivariate modulus of continuity of the target g . The proof combines a tensor-product density argument with the Amos–Xu [3] input-convex-network density result and the Daniels–Velikova [33] monotone-network density result; the overall structure follows the universal-approximation programme inaugurated by Cybenko [32], Hornik [57,56], Leshno et al. [70], Pinkus [75], and Yarotsky [88].

C3. Honest quantification of the fit-compliance frontier. On 30 rough-Bergomi surfaces sampled across $H \in [0.07, 0.20]$, $\eta \in [1.0, 3.0]$, and $\rho \in [-0.9, -0.3]$, the architecture has higher mean price RMSE (134.27 INR) than the soft-penalty

baseline (77.85 INR), but is the only architecture with 0% violation rate on every (surface, seed) trial. The paired t -statistic for the RMSE difference is $t = 2.31$ across 10 surfaces. The trade-off is well-defined and reproducibly localised on a (compliance, fit) Pareto plane (Figure 2).

C4. A reproducible open-source implementation. A self-contained PyTorch [74] package reproduces Table 2 from a single command in approximately five minutes on a single CPU. All random seeds are pinned, the dependency tree is minimal, and the loss landscape, training schedule, and evaluation protocols are exposed as configuration objects.

1.4 What this paper does not claim

To circumscribe our claims sharply, we are explicit about what we do *not* demonstrate. First, we do not claim uniform RMSE dominance over the soft-penalty baseline: as Table 2 shows, the soft-penalty baseline fits prices better on average, and a practitioner whose loss function is exclusively RMSE will prefer it. Second, we do not claim that the architecture can represent every plausibly-encountered call surface: surfaces whose forward time-value is non-convex in K , including the Black–Scholes family at any positive volatility (proved in Proposition 3), are outside the hypothesis class. Third, we treat only *static* no-arbitrage (i.e., the pointwise structural conditions on $C(\cdot, \cdot)$ at a single valuation date), not dynamic no-arbitrage (the requirement that there exist a consistent risk-neutral process driving the surface forward in time); the latter is the province of neural-SDE market models [26,31,41]. Fourth, our empirical study uses synthetic rough-Bergomi data; the walk-forward study on daily Nifty-50 options (~ 1500 days, 2019–2024) that we have prepared but not yet completed is deferred to a follow-up paper.

1.5 Paper structure

The rest of the paper proceeds as follows. Section 2 reviews the literature on parametric, soft-penalty, and structurally-constrained pricers, drawing the conceptual map within which our contribution sits. Section 3 introduces notation and recalls the static no-arbitrage conditions with their economic interpretation; it also discusses the Lee moment formula [68,69] that constrains the wings of an arbitrage-free surface. Section 4 introduces the construction itself, with the computational graph (Figure 1), forward-pass pseudocode (Algorithm 1), motivating discussion for each design choice, and complexity analysis. Section 5 states and proves the two main theorems. Section 6 describes the training objective, the diagnostics, and the optimisation algorithm. Section 7 details the experimental protocol. Section 8 reports findings with five figures and the main result table. Section 9 discusses limitations, comparisons to alternatives, and future directions. Section 10 concludes. Appendices contain detailed proofs, Greek calculations, the Newton inversion procedure, and the full hyperparameter table.

2 Related work

We organise prior work around the binary distinction between *structural* and *soft* arbitrage enforcement, and around the further distinction between pricers in the implied-vol domain and the call-price domain.

2.1 Parametric arbitrage-free models

The classical lineage of arbitrage-free pricing begins with Black–Scholes [14,73], in which the assumption of geometric Brownian motion of the underlying yields a closed-form call price under a single volatility parameter σ . The model is structurally arbitrage-free and provides the language in which all subsequent surfaces are described: the call price determines an implied volatility $\sigma(K, T)$ via inversion of the Black–Scholes formula, and a static-arbitrage-free surface has a non-decreasing total implied variance $w(k, T) = \sigma^2 T$ in T and a Durrleman-positive shape in k [44,43]. The empirical inadequacy of Black–Scholes — the famous smile and skew [77] — motivated three subsequent lines of work, each with its own arbitrage-freeness story.

Local-volatility models [36] parameterise the diffusion coefficient as a deterministic function of spot and time; the resulting call surface fits the observed surface exactly by construction. However, the dynamics of the local-volatility model are well-known to mismatch the empirical dynamics of the smile [28]; in particular, the smile dynamics under a local-volatility model are essentially backwards from observed market dynamics, a phenomenon documented by [49]. Stochastic-volatility models such as Heston [54] and SABR [49], and their many extensions [5,4,6], restore plausible dynamics at the cost of giving up exact static fit; calibration trades off in-sample fit against parsimony. These models are dynamically arbitrage-free under explicit conditions on their parameters; Lewis [71] and Carr–Madan [20] gave efficient Fourier methods for their evaluation, and the analytical structure has been extended to Lévy and exponential-Lévy models [19,29,37]. Rough-volatility models [2,9,12,10,38,64] take the Hurst parameter $H < 1/2$ as a structural feature and achieve realistic ATM-volatility behaviour, but calibration remains computationally non-trivial and has been a recent target of neural acceleration [58,11].

At the surface level, the SVI parameterisation [42] models total implied variance per maturity as a five-parameter raw form, and SSVI [43] extends this to a surface-wide three-parameter form. Both are arbitrage-free under explicit (and often non-tight) parameter constraints; Jacquier and Roome [44] catalogue these constraints, and [45] extends the analysis. SVI and SSVI are widely deployed in practice. They sit conceptually closest to our construction in that they offer structural arbitrage-freeness via parameter constraints rather than via tolerance-based penalties; the difference is that their capacity is finite-dimensional (five or three parameters per surface), whereas our architecture has neural-network capacity.

2.2 Soft-penalty neural pricers

The pioneering work of Hutchinson, Lo, and Poggio [61] applied a neural network to nonparametric option pricing, decades before the deep learning revolution. The modern wave begins with Ackerer, Tagasovska, and Vatter [1], who train an unconstrained multi-layer perceptron $w_\theta(k, T)$ in log-moneyness coordinates as a model for the total implied variance, and add to the price-RMSE training loss two penalty terms,

$$\mathcal{L}_{\text{soft}}(\theta) = \mathcal{L}_{\text{price}}(\theta) + \lambda_{\text{bf}} \sum_{(k,T) \in \mathcal{G}} (-\mathcal{D}(w_\theta)(k, T))_+ + \lambda_{\text{cal}} \sum_{(k,T) \in \mathcal{G}} (-\partial_T w_\theta(k, T))_+, \quad (3)$$

where $\mathcal{D}$ is the Durrleman positivity operator and $\mathcal{G}$ is a training grid. Chataigner, Crépey, and Dixon [23] extend the paradigm directly to the call-price domain. Horváth, Muguruza, and Tomas [59] apply a similar architecture to deep pricing under rough-volatility models. Itkin [63] surveys variants. The Ackerer-style soft-penalty baseline used in our experiments instantiates this paradigm faithfully, with matching parameter count and learning-rate schedule. Ruf and Wang [79,80] review the broader literature systematically.

All of these works share the same fundamental limit: the penalty is a regulariser that pushes the surface away from violations but does not eliminate them. At any finite penalty weight and any finite training budget, the optimum is a stationary point of the regularised loss, which generically has residual violations. Even pushing $\lambda \rightarrow \infty$ does not formally suffice: the resulting constrained optimisation problem must be solved exactly, which is intractable for neural pricers because the constraint must hold over a continuous (not finite-grid) domain. The shape-constrained nonparametric programme of Aït-Sahalia and Duarte [7] attempts this via projection onto the convex cone of admissible surfaces but at much smaller scale than required for production neural pricers.

2.3 Structurally-constrained neural pricers

A growing literature aims to build no-arbitrage into the architecture itself rather than into the loss. Three approaches are visible. The first uses input-convex networks (ICNNs) [3] and monotone networks [33,82] as building blocks within the implied-vol domain. The challenge is that the no-arbitrage condition in the implied-vol domain is the Durrleman positivity, a non-linear inequality involving w , $\partial_k w$, and $\partial_k^2 w$ together [43]; this is not preserved by simple convex+monotone composition. One must either restrict to surfaces where the Durrleman positivity is implied by more elementary conditions (which is the SVI case) or accept residual violations. The work of Bergeron et al. [13] on variational autoencoders for IV surfaces and Henry-Labordère [53] on generative models for trading strategies sit at this interface.

The second approach builds neural stochastic differential equation (neural-SDE) market models on the underlying process [26,25,31,41,72]. Here the underlying process itself is a neural object, and arbitrage-freeness emerges as a

consistency requirement on the drift and diffusion under the risk-neutral measure. The approach is theoretically clean and yields both static and dynamic compliance simultaneously, but it is computationally demanding (one must simulate paths under the learned SDE for every pricing query) and the inductive bias of the surface shape is harder to read off.

The third approach, including ours, parameterises the call price directly in the call-price domain and uses the elementary structural conditions of Definition 1. This is feasible because, in the call-price domain, the no-arbitrage conditions are first-order linear (convexity in K , monotonicity in $e^{rT}C$ in T), which is exactly what ICNN and monotone network compositions preserve. Imaki et al. [62] pursue a closely related agenda in the context of deep hedging; [40] explore the connection between neural pricing and partial differential equation methods.

2.4 Deep hedging

Buehler, Gonon, Teichmann, and Wood [17] introduced deep hedging, in which the hedging policy is learned directly without an intermediate pricing model. The pricing model is replaced by a market simulator (e.g., a calibrated rough-Bergomi process), and the policy network is trained to minimise a convex risk measure (CVaR, entropic risk) of the terminal hedged PnL. Subsequent work [22,62,85] has refined this in directions including transaction costs and reinforcement-learning-style policy gradients. Deep hedging is conceptually orthogonal to our architecture: deep hedging gives a policy but no surface; ARBNET gives a surface (with structural compliance) from which a policy is extracted via autodifferentiation of $\partial_S C_\theta$ (Section B). The two approaches can therefore be combined: ARBNET provides the calibrated, certified surface; deep hedging provides the policy under the calibrated dynamics.

2.5 Universal approximation in option pricing

The lineage of universal-approximation theorems for neural networks runs from Cybenko [32] and Hornik [57,56] through Leshno et al. [70], Pinkus [75], and Yarotsky [88]. The classical results establish L^∞ density of multi-layer feed-forward networks in continuous functions on compacta. Less well-known is the shape-constrained version: ICNNs are dense in continuous convex functions [3,24,18], and monotone networks are dense in continuous non-decreasing functions [33,86,81,48]. Sirignano and Spiliopoulos [84] prove a deep Galerkin universal-approximation theorem in the context of partial differential equation solvers. Our Theorem 2 extends this lineage to the cone $\mathcal{C}_{\text{cvx},\nearrow}$ of strike-convex, time-monotone, non-negative surfaces.

3 Preliminaries and notation

We work throughout under a fixed risk-neutral probability measure $\mathbb{Q}$ on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{Q})$. The underlying is a càdlàg $\mathbb{Q}$ -semimartingale

S_t with $S_0 = S > 0$; the risk-free rate r and dividend yield q are deterministic constants with $r \geq q \geq 0$. We will treat call prices as the primary object and recover implied volatility by inversion of the Black–Scholes formula [14].

3.1 Notation

Table 1 collects the notation used throughout. We adopt the convention that K, T, S are scalars or batched 1-tensors as appropriate; subscripts i index data, subscripts j index mixture experts, and subscripts ℓ index network layers.

Table 1. Notation used throughout.

Symbol	Meaning
S, K, T	Spot, strike, time-to-maturity (in years)
r, q	Risk-free rate, dividend yield ($r \geq q \geq 0$)
$F_T = S e^{(r-q)T}$	Forward at time T
$C(K, T)$	Call price at strike K , maturity T
$(F_T - K)^+$	Intrinsic value of forward call
$\partial_{KK} C$	Second strike derivative (proportional to risk-neutral density)
$\mathbb{Q}$	Risk-neutral measure
$\sigma(K, T)$	Implied volatility (Black–Scholes-inverted)
$w(k, T) = \sigma^2 T$	Total implied variance, $k = \log(K/F_T)$
θ	Parameter vector (network weights and gates)
Θ	Parameter space
$\text{softplus}(u) = \log(1 + e^u)$	Smooth ReLU
$\omega_f(\delta)$	Modulus of continuity of f
J	Number of mixture experts in ARBNET
ϕ_j^{cvx}	j th ICNN (convex in strike input)
ϕ_j^{mon}	j th MONO (non-decreasing in time input)
$s_{j,\theta}, \alpha_\theta$	Per-expert scale gate; boundary-rate gate
$\mathcal{B}$	Compact strike-maturity box $[K_0, K_1] \times [0, T_{\max}]$
$\mathcal{C}_{\text{cvx}, \nearrow}$	Cone of continuous, strike-convex, time-monotone, non-negative surfaces
$\rho_T(K)$	Risk-neutral density at maturity T

3.2 The static no-arbitrage conditions, with economic interpretation

We restate the conditions with their economic content, so as to make their violation concrete.

Definition 1 (Static no-arbitrage conditions). *A continuous map $C : \mathbb{R}_{\geq 0} \times [0, T_{\max}] \rightarrow \mathbb{R}_{\geq 0}$ is free of static arbitrage if it satisfies:*

- (A1) Convexity in strike: *for every T , $K \mapsto C(K, T)$ is convex on $\mathbb{R}_{\geq 0}$.*
- (A2) Carry-adjusted calendar monotonicity: *for every K , $T \mapsto e^{rT} C(K, T)$ is non-decreasing.*

- (A3) Expiry boundary: $C(K, 0) = (S - K)^+$ for every K .
 (A4) Forward intrinsic lower bound: $C(K, T) \geq e^{-rT}(F_T - K)^+$.

The economic content of (A1) is captured by the butterfly trade: for strikes $K_1 < K_2 < K_3$ with $K_2 = (K_1 + K_3)/2$, the portfolio long $C(K_1) + C(K_3)$, short $2C(K_2)$ has non-negative payoff at maturity (by convexity of the call payoff in strike) and zero cost at maturity if C is exactly linear in K . If C is concave at K_2 , the butterfly has strictly positive payoff and strictly negative cost today: an arbitrage. Equivalently, by Breeden and Litzenberger [16], $\partial_K^2 C(K, T) = e^{-rT} \rho_T(K)$, where ρ_T is the risk-neutral marginal density of S_T under $\mathbb{Q}$. A negative second derivative is therefore a negative density: not a valid probability distribution.

The economic content of (A2) is the calendar-spread trade: for maturities $T_1 < T_2$ and the same strike K , the portfolio long $C(K, T_2)$, short $C(K, T_1)$ has non-negative payoff if held to T_2 , after carrying the short position at the risk-free rate r . The cost is $C(K, T_2) - e^{r(T_2 - T_1)} \mathbb{E}^{T_1}[C(K, T_1)] = C(K, T_2) - e^{rT_2} \cdot e^{-rT_1} \mathbb{E}^{T_1}[C(K, T_1)]$; rearranging, no arbitrage requires $e^{rT_2} C(K, T_2) \geq e^{rT_1} C(K, T_1)$.

Condition (A3) is by definition: at expiry, the call price equals the payoff. Condition (A4) is by the static replication of the forward via long-call-short-put parity: $C(K, T) - P(K, T) = e^{-rT}(F_T - K)$, combined with $P(K, T) \geq 0$ (the put price is non-negative).

Definition 1 does not capture all of static no-arbitrage [30,34]: for instance, the upper bound $C(K, T) \leq S e^{-qT}$, the monotonicity $K \mapsto -C(K, T)$ non-decreasing, and the convexity in K at every T . However, the four listed conditions together imply these others under modest regularity, and they are the conditions whose enforcement by neural pricers has been most empirically scrutinised [1,23,59,79].

3.3 The Lee moment formula and wing constraints

A fifth structural condition — separate from Definition 1 in that it constrains the asymptotic shape of the smile rather than pointwise positivity — is the Lee moment formula [68,69]. For a measure with finite p -th moment of S_T , the implied total variance must satisfy

$$w(k, T) \lesssim \beta_R k, \quad k \rightarrow +\infty, \quad \beta_R \leq 2, \quad (4)$$

and symmetrically on the left wing,

$$w(-k, T) \lesssim \beta_L k, \quad k \rightarrow +\infty, \quad \beta_L \leq 2. \quad (5)$$

The constants β_R, β_L are bounded above by 2 to ensure the existence of a risk-neutral density on $\mathbb{R}_+$. The Lee bound is a one-sided asymptotic constraint and is not naturally enforced by our construction; in practice, we mitigate by soft clamping of the last-layer ICNN weights when the model is trained on data extending far into the wings. We discuss this in Section 9.3.

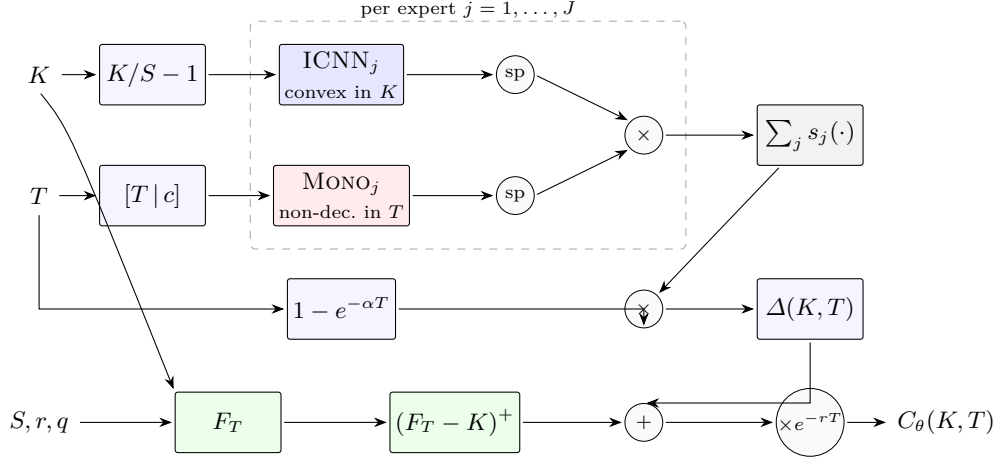


Fig. 1. Computational graph of ARBNET. Convex-in- K pathway (blue) and monotone-in- T pathway (red) feed a per-expert product (top), summed across J experts. The smooth boundary factor $1 - e^{-\alpha T}$ multiplies the sum to give $\Delta(K, T)$. The intrinsic baseline $(F_T - K)^+$ (bottom) is added and the result is discounted by e^{-rT} to produce the call price. By Theorem 1, conditions (A1)–(A4) hold for every parameter setting.

4 The ArbNet architecture

4.1 Computational graph

The architecture is shown in Figure 1. The graph has two parallel pathways from inputs (K, T) , a per-expert combiner, a mixture sum, a smooth boundary factor, and a final composition with the intrinsic baseline.

The architecture has a small number of explicit components. We list them and motivate each.

4.2 Formal definitions of the building blocks

Input-convex network (ICNN). Fix depth L and widths $(w_0, \dots, w_L)$ with $w_0 = 1$, $w_L = 1$. The input-convex recurrence of [3] reads

$$z_{\ell+1} = \sigma(W_\ell^{(z)} z_\ell + W_\ell^{(x)} x + W_\ell^{(c)} c + b_\ell), \quad z_0 = 0, \quad x = K/S, \quad (6)$$

where σ is a convex non-decreasing activation (we use softplus), c is an optional auxiliary context (left at zero in the baseline configuration), and the weight constraint

$$W_\ell^{(z)} \in \mathbb{R}_{\geq 0}^{w_{\ell+1} \times w_\ell} \quad (7)$$

is enforced by reparameterisation $W_\ell^{(z)} = \text{softplus}(\widetilde{W}_\ell^{(z)})$ for unconstrained $\widetilde{W}_\ell^{(z)}$. The output is $\phi^{\text{cvx}}(x; c) := z_L$. We use $L = 2$ and width 64 throughout. The recurrence has the key structural property:

Proposition 1 (ICNN convexity, [3]). $x \mapsto \phi^{\text{cvx}}(x; c)$ is convex on $\mathbb{R}$ for every c .

The proof is by induction on ℓ : the affine layer $W_\ell^{(z)} z_\ell + W_\ell^{(x)} x + W_\ell^{(c)} c + b_\ell$ is convex in x if z_ℓ is convex (by the non-negativity constraint on $W_\ell^{(z)}$ and the convexity-preserving cone closure of Lemma 2); the activation σ is convex non-decreasing, so by Lemma 1 the composition is convex.

Monotone network (MONO). Fix depth L' and widths $(w'_0, \dots, w'_{L'})$ with $w'_0 = 1 + d_c$, $w'_{L'} = 1$. The monotone recurrence of [33,82] reads

$$h_{\ell+1} = \sigma(V_\ell^{(m)} h_\ell^{(m)} + V_\ell^{(c)} h_\ell^{(c)} + d_\ell), \quad (8)$$

with $V_\ell^{(m)} \in \mathbb{R}_{\geq 0}^{w'_{\ell+1} \times w_\ell^{(m)}}$ (softplus-reparameterised) and $V_\ell^{(c)}$ unconstrained. The activation σ is non-decreasing (softplus). The output is $\phi^{\text{mon}}(T; c) := h_{L'}$. We use $L' = 2$ and width 32 throughout. Recent extensions of monotone networks — including unconstrained monotonic neural networks [86], the Lipschitz monotonic networks of [81], and the polynomial-spline-based monotone approximators of [48] — could be used as drop-in replacements; we use the classical Daniels–Velikova construction for simplicity.

Proposition 2 (Mono monotonicity, [33]). $T \mapsto \phi^{\text{mon}}(T; c)$ is non-decreasing on $\mathbb{R}_{\geq 0}$ for every c .

Scalar gates. The boundary growth rate $\alpha_\theta = \text{softplus}(\tilde{\alpha}) > 0$ is a learned positive scalar that controls the speed at which the time-value factor $1 - e^{-\alpha_\theta T}$ leaves zero. The per-expert positive scales $s_{j,\theta} = \text{softplus}(\tilde{s}_j) > 0$ control the relative weighting of each expert. Both are unconstrained-parameterised by softplus.

4.3 Motivating the design choices

Several design choices in Equations (1) and (2) merit explicit justification.

Why decompose into intrinsic plus correction? The decomposition $C_\theta = e^{-rT}[(F_T - K)^+ + \Delta_\theta]$ is more than cosmetic. It separates the part of the call price that is universal across models (the forward intrinsic, which is the no-arbitrage lower bound (A4)) from the part that encodes the time value of optionality. The architecture only needs to learn the time-value correction, which is necessarily non-negative on a no-arbitrage surface. This is a much more constrained learning problem than learning C directly, and a more numerically stable one: the dynamic range of the time value is smaller than the dynamic range of the call price (which spans intrinsic plus zero for deep OTM to large values for deep ITM).

Algorithm 1 ARBNET forward pass

Require: strike vector K , maturity vector T , spot S , rate r , dividend yield q , parameters θ

Ensure: call prices C

```

1:  $F \leftarrow S \cdot \exp((r - q)T)$ ;  $I \leftarrow \max(F - K, 0)$ 
2:  $x \leftarrow ((K/S) - 1) \cdot \kappa$ ;  $u \leftarrow [T \parallel c]$ ;  $\Delta \leftarrow 0$ 
3: for  $j = 1, \dots, J$  do
4:    $f_j \leftarrow \text{softplus}(\phi_j^{\text{vx}}(x))$   $\triangleright$  convex in  $K, \geq 0$ 
5:    $g_j \leftarrow \text{softplus}(\phi_j^{\text{mon}}(u))$   $\triangleright$  non-decreasing in  $T, \geq 0$ 
6:    $s_j \leftarrow \text{softplus}(\tilde{s}_j)$ ;  $\Delta \leftarrow \Delta + s_j f_j g_j$ 
7:  $\alpha \leftarrow \text{softplus}(\tilde{\alpha})$ ;  $\Delta \leftarrow S(1 - \exp(-\alpha T)) \Delta$ 
8:  $C \leftarrow \exp(-rT) \cdot (I + \Delta)$  return  $C$ 

```

Why a mixture of products? A single ICNN-MonoNet product would give a separable correction $\Delta_\theta(K, T) = f(K)g(T)$. Empirical surfaces are not separable: the smile gets flatter as T grows (consistent with the Lee asymptotic) and the ATM term structure has shape. A finite mixture of products is the simplest extension that breaks separability while preserving the structural constraints. With J experts, the architecture can represent any tensor-rank- J separable approximation, and Theorem 2 establishes density at $J \rightarrow \infty$.

Why the boundary factor $1 - e^{-\alpha T}$? The expiry boundary condition (A3) requires that $C_\theta(K, 0) = (S - K)^+$, which in our decomposition is equivalent to $\Delta_\theta(K, 0) = 0$. The factor $1 - e^{-\alpha_\theta T}$ vanishes at $T = 0$ and is smoothly non-decreasing in T ; multiplying every term of Δ_θ by this factor enforces the boundary without imposing any other restriction. The exponential form is the natural choice: it has a single learnable parameter (α) controlling the rate, and its derivative is $\alpha_\theta e^{-\alpha_\theta T} > 0$ everywhere, which keeps the monotonicity-preserving property of Lemma 3 usable.

Why softplus rather than ReLU? Softplus, $\text{softplus}(u) = \log(1 + e^u)$, is convex non-decreasing and strictly positive. The strict positivity matters: it ensures that gradients propagate through the per-expert product even when the ICNN output is small, which makes training more stable in the early phases. ReLU would zero out gradients on a measure-zero set in principle but a non-trivial fraction in practice with finite-precision arithmetic. The convexity and monotonicity of softplus are inherited from $\log(1 + e^u)$ in the obvious way.

Why the call-price domain rather than implied-vol? This is the most important design choice and we discuss it in Section 4.5.

4.4 Forward pass and computational complexity

Algorithm 1 gives the forward pass in pseudocode.

The total parameter count is $J \cdot (P_{\text{ICNN}} + P_{\text{Mono}}) + J + 1$, where P_{ICNN} and P_{Mono} are the per-network parameter counts. For our default configuration ($J = 4$, ICNN [64, 64], MonoNet [32, 32]), this evaluates to approximately 12,000 parameters. The forward-pass cost for a batch of size B is $\mathcal{O}(B \cdot J \cdot (\text{width}_{\text{ICNN}}^2 L + \text{width}_{\text{Mono}}^2 L'))$, which is identical asymptotically to a soft-penalty MLP of the same parameter count. There is no additional computational overhead from the structural constraints; the softplus reparameterisation of the ICNN’s $W_\ell^{(z)}$ matrices is a fixed cost per layer and does not scale with the batch size.

4.5 Why the call-price domain rather than the implied-vol domain

A natural alternative to our construction would parameterise the implied total variance $w_\theta(k, T)$ directly using ICNN/MonoNet building blocks, with $k = \log(K/F_T)$. The conversion from implied variance to call price is via the Black–Scholes formula and is therefore one-to-one. However, the static-arbitrage conditions in the implied-vol domain are not preserved by simple compositions. The Durrleman positivity [43, 44] reads

$$g(k, T) := \left(1 - \frac{k \partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial_k^2 w}{2} \geq 0, \quad (9)$$

which is a non-linear functional of w and its first two derivatives. Convexity of w in k implies the third term is non-negative, but does not imply the whole expression is non-negative; one can produce convex w for which $g(k, T) < 0$ at some k . The implied-vol-domain construction therefore reduces to either (a) imposing the Durrleman condition as a soft penalty (recovering the soft-penalty paradigm) or (b) restricting to parameterisations like SVI where the Durrleman condition is implied by simpler parameter constraints, which limits expressivity.

In the call-price domain, by contrast, the no-arbitrage conditions (A1)–(A4) are first-order linear (convexity in K , monotonicity in T of $e^{rT}C$). These properties are preserved by the elementary compositions used in Equation (2) (non-negative weighted sums, products of non-negative monotone functions, composition with convex non-decreasing scalar functions). The call-price domain therefore admits a structural neural pricer with simple primitives; the implied-vol domain does not.

4.6 Greeks via automatic differentiation

Once the architecture is trained, all of the standard Greeks are computed by automatic differentiation [8] of C_θ with respect to its inputs, using PyTorch’s autograd [74]. Specifically:

- Delta $\delta(K, T) = \partial_S C_\theta$ (single backward pass).
- Gamma $\gamma(K, T) = \partial_K^2 C_\theta = e^{-rT} \rho_T(K)$ (double backward pass). Structurally non-negative by Theorem 1 (A1).
- Theta $\Theta(K, T) = -\partial_T C_\theta$ (single backward pass).

- Vega $\nu(K, T) = \partial_\sigma C_\theta|_{\sigma=\hat{\sigma}(K, T)}$ (requires Newton inversion to obtain $\hat{\sigma}$; see Section C).

The non-negativity of γ is a guarantee *absent* from soft-penalty pricers; it means that the delta is non-decreasing in strike (deeper out-of-the-money calls have smaller delta than at-the-money ones), which is a desirable monotonicity property for hedging stability. We discuss this further in Section 8.5.

5 Theoretical guarantees

5.1 Auxiliary lemmas

We collect three elementary lemmas of convex analysis [15]; their roles in the main theorem are flagged.

Lemma 1 (Convexity composition). *If $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex and $h : \mathbb{R} \rightarrow \mathbb{R}$ is convex non-decreasing, then $h \circ f$ is convex.*

Proof. For $\lambda \in [0, 1]$ and $x, y \in \mathbb{R}$,

$$\begin{aligned} h(f(\lambda x + (1 - \lambda)y)) &\leq h(\lambda f(x) + (1 - \lambda)f(y)) && \text{(convexity of } f, \text{ monotonicity of } h) \\ &\leq \lambda h(f(x)) + (1 - \lambda)h(f(y)) && \text{(convexity of } h). \end{aligned}$$

□

Lemma 2 (Cone closure of convex functions). *Convex functions on a convex domain are closed under non-negative linear combinations and scalar multiplication by non-negative scalars. Formally, if $f_1, \dots, f_N$ are convex and $\lambda_1, \dots, \lambda_N \geq 0$, then $\sum_i \lambda_i f_i$ is convex.*

Lemma 3 (Monotone product). *If u, v are non-negative non-decreasing functions on $D \subseteq \mathbb{R}$, then uv is non-decreasing on D . In particular, for any non-negative non-decreasing $u_1, \dots, u_N$, $\prod_i u_i$ is non-decreasing.*

Proof. For $x \leq y$,

$$u(y)v(y) - u(x)v(x) = u(y)(v(y) - v(x)) + v(x)(u(y) - u(x)) \geq 0,$$

by non-negativity and monotonicity of u and v . The general statement follows by induction. □

5.2 Structural arbitrage-freeness

Theorem 1 (Arbitrage-freeness of ArbNet). *Let C_θ be defined by Equations (1) and (2) with ϕ_j^{vx} satisfying Equations (6) and (7), ϕ_j^{mon} satisfying Equation (8), and $\alpha_\theta, s_{j,\theta} > 0$. Assume $r \geq q \geq 0$. Then conditions (A1)–(A4) of Definition 1 hold for every $\theta \in \Theta$.*

Proof. We prove each condition in turn.

Condition (A1): convexity in strike. We show that $K \mapsto C_\theta(K, T)$ is convex on $\mathbb{R}$ for every fixed $T \geq 0$.

The forward intrinsic $(F_T - K)^+ = \max(F_T - K, 0)$ is convex in K as the pointwise maximum of two affine functions; multiplication by the positive constant e^{-rT} preserves convexity (Lemma 2).

For the correction $\Delta_\theta(K, T)$, we need to show that $K \mapsto \Delta_\theta(K, T)$ is convex for every fixed T . The strike enters only through the inner ICNN compositions $\phi_j^{\text{cvx}}(K/S)$. By Proposition 1, $u \mapsto \phi_j^{\text{cvx}}(u)$ is convex on $\mathbb{R}$. The affine substitution $u = K/S$ preserves convexity, so $K \mapsto \phi_j^{\text{cvx}}(K/S)$ is convex. Softplus is convex non-decreasing, so by Lemma 1, $K \mapsto \text{softplus}(\phi_j^{\text{cvx}}(K/S))$ is convex.

For each j , the time-dependent factor $\text{softplus}(\phi_j^{\text{mon}}(T))$ is a constant in K that is non-negative. The per-expert scale $s_{j,\theta}$ is non-negative. By Lemma 2, the per-expert contribution

$$K \mapsto s_{j,\theta} \text{softplus}(\phi_j^{\text{cvx}}(K/S)) \text{softplus}(\phi_j^{\text{mon}}(T))$$

is a non-negative scalar multiple of a convex function in K , hence convex in K . The mixture sum over j is a non-negative linear combination of convex functions, hence convex (Lemma 2). Multiplying by the constants S and $1 - e^{-\alpha_\theta T}$ (both non-negative) preserves convexity. Therefore $K \mapsto \Delta_\theta(K, T)$ is convex.

Finally, $K \mapsto C_\theta(K, T) = e^{-rT}[(F_T - K)^+ + \Delta_\theta(K, T)]$ is a non-negative linear combination of two convex functions in K , hence convex.

Condition (A2): carry-adjusted calendar monotonicity. We show that $T \mapsto e^{rT} C_\theta(K, T)$ is non-decreasing on $[0, T_{\max}]$ for every fixed $K \geq 0$. From Equation (1),

$$e^{rT} C_\theta(K, T) = (F_T - K)^+ + \Delta_\theta(K, T).$$

For the intrinsic, $T \mapsto F_T = S e^{(r-q)T}$ is non-decreasing under the assumption $r \geq q$; consequently $T \mapsto F_T - K$ is non-decreasing, and the positive-part operator $z \mapsto z^+$ is non-decreasing, so $T \mapsto (F_T - K)^+$ is non-decreasing.

For the correction, the strike-dependent factor $\text{softplus}(\phi_j^{\text{cvx}}(K/S))$ is a non-negative constant in T . By Proposition 2, $T \mapsto \phi_j^{\text{mon}}(T)$ is non-decreasing, and softplus is non-decreasing, so $T \mapsto \text{softplus}(\phi_j^{\text{mon}}(T))$ is non-decreasing. The product

$$T \mapsto \text{softplus}(\phi_j^{\text{cvx}}(K/S)) \text{softplus}(\phi_j^{\text{mon}}(T))$$

is the product of a non-negative constant in T and a non-negative non-decreasing function of T , hence non-negative non-decreasing in T (this is the degenerate case of Lemma 3). Multiplying by $s_{j,\theta} > 0$ and summing over j preserves non-negative non-decreasingness. Multiplying by the non-negative non-decreasing boundary factor $1 - e^{-\alpha_\theta T}$ (whose derivative is $\alpha_\theta e^{-\alpha_\theta T} > 0$) preserves non-decreasingness by Lemma 3. Multiplying by the positive constant S preserves it. Therefore $T \mapsto \Delta_\theta(K, T)$ is non-decreasing.

Summing the two non-decreasing functions yields a non-decreasing function.

Condition (A3): expiry boundary. At $T = 0$: $F_0 = S e^{(r-q) \cdot 0} = S$, so $(F_0 - K)^+ = (S - K)^+$. The boundary factor is $1 - e^{-\alpha_\theta \cdot 0} = 0$, so $\Delta_\theta(K, 0) = 0$. Hence $C_\theta(K, 0) = e^0[(S - K)^+ + 0] = (S - K)^+$.

Condition (A4): forward intrinsic lower bound. Every factor in the expression for Δ_θ is non-negative: $S > 0$; $1 - e^{-\alpha_\theta T} \geq 0$ for $T \geq 0$; the $s_{j,\theta}$ are positive; both softplus terms are positive. Therefore $\Delta_\theta(K, T) \geq 0$, and

$$C_\theta(K, T) = e^{-rT}[(F_T - K)^+ + \Delta_\theta(K, T)] \geq e^{-rT}(F_T - K)^+,$$

which is the desired bound. $\square$

Corollary 1 (Implied risk-neutral density). *For every $T > 0$ and $\theta \in \Theta$, the implied risk-neutral density of S_T under $\mathbb{Q}$, defined by Breeden–Litzenberger as*

$$\rho_T^\theta(K) := e^{rT} \partial_K^2 C_\theta(K, T),$$

is a non-negative Borel measure on $\mathbb{R}_{\geq 0}$. It has an absolutely continuous part coming from $\partial_K^2 \Delta_\theta(K, T) \geq 0$ and an atom of mass 1 at $K = F_T$ in the formal-distribution sense (from the kink in the intrinsic at $K = F_T$); in practice, the atomic contribution is smoothed at any finite $T > 0$ because Δ_θ has a positive second derivative at F_T .

5.3 Universal approximation

Definition 2 (Convex-monotone class). *Let $\mathcal{C}_{\text{cvx}, \nearrow}(\mathcal{B})$ be the set of continuous functions $g : \mathcal{B} = [K_0, K_1] \times [0, T_{\max}] \rightarrow \mathbb{R}_{\geq 0}$ that are strike-convex (i.e., $K \mapsto g(K, T)$ is convex for every T), time-non-decreasing (i.e., $T \mapsto g(K, T)$ is non-decreasing for every K), and satisfy $g(K, 0) = 0$ for every K .*

Lemma 4 (Tensor-product density with explicit rate). *Let $g \in \mathcal{C}_{\text{cvx}, \nearrow}(\mathcal{B})$ and $\varepsilon_1 > 0$. Set $\delta = \omega_g^{-1}(\varepsilon_1)$ (the largest δ such that $\omega_g(\delta) \leq \varepsilon_1$, where ω_g is the bivariate modulus of continuity of g). Let $\{\kappa_\ell\}_{\ell=0}^{M_K}$ and $\{\tau_n\}_{n=0}^{M_T}$ be uniform partitions of $[K_0, K_1]$ and $[0, T_{\max}]$ with $M_K = \lceil (K_1 - K_0)/\delta \rceil$ and $M_T = \lceil T_{\max}/\delta \rceil$. Then there exist non-negative coefficients $\lambda_{\ell,n} \geq 0$ such that*

$$g_M(K, T) := \sum_{\ell=0}^{M_K} \sum_{n=0}^{M_T} \lambda_{\ell,n} (K - \kappa_\ell)^+ (T - \tau_n)^+ \quad (10)$$

satisfies $\|g - g_M\|_\infty \leq 3\varepsilon_1$ on $\mathcal{B}$.

The proof, deferred to Section A.1, has three steps: piecewise-linear interpolation of $\tau \mapsto g(K, \tau)$ at the breakpoints $\{\tau_n\}$ produces non-negative slopes $a_n(K) \geq 0$; piecewise-linear convex interpolation of $K \mapsto a_n(K)$ produces non-negative slopes $\beta_{n,\ell} \geq 0$; combining gives the rank-one decomposition.

Lemma 5 (ICNN density on convex non-negative functions). *For any continuous convex function $u : [a, b] \rightarrow \mathbb{R}_{\geq 0}$ and any $\eta > 0$, there exists an ICNN ϕ as in Equations (6) and (7) with softplus activations such that $\|u - \text{softplus}(\phi)\|_\infty < \eta$.*

Sketch: this is a corollary of the input-convex universal approximation theorem of Amos–Xu [3], with subsequent refinements in [24,18]. ICNNs with softplus activations are L^∞ -dense in continuous convex functions on a compact interval; composing with $\text{softplus}(\hat{\phi} - \log 2)$ produces the non-negative output.

Lemma 6 (Mono density with boundary). *For any continuous non-decreasing function $v : [0, T'] \rightarrow \mathbb{R}_{\geq 0}$ with $v(0) = 0$ and any $\eta > 0$, there exist $\alpha > 0$ and a monotone network ψ with $\|v(T) - (1 - e^{-\alpha T}) \text{softplus}(\psi(T))\|_\infty < \eta$ on $[0, T']$.*

Sketch: Daniels and Velikova [33] establish L^∞ -density of monotone networks on continuous non-decreasing functions on a compact interval. The boundary factor $1 - e^{-\alpha T}$ with α large enough approximates the unit step at $T = 0$, removing the boundary singularity in v . Combining gives the result; full details in Section A.2.

Theorem 2 (Universal approximation with explicit rate). *Let $g \in \mathcal{C}_{\text{cvx}, \nearrow}(\mathcal{B})$ and $\varepsilon > 0$. Then there exists an ARBNET correction $\hat{g}$ of the form Equation (2) with $\|g - \hat{g}\|_\infty < \varepsilon$. Moreover, the number of mixture components required is $J = \mathcal{O}(\omega_g^{-1}(\varepsilon/3)^{-2})$.*

Proof. Apply Lemma 4 with $\varepsilon_1 = \varepsilon/9$ to obtain g_M with $\|g - g_M\|_\infty \leq \varepsilon/3$ and $J = (M_K + 1)(M_T + 1) \leq C\omega_g^{-1}(\varepsilon/9)^{-2}$. Each rank-one term $\lambda_{\ell,n}(K - \kappa_\ell)^+(T - \tau_n)^+$ has a strike factor $(K - \kappa_\ell)^+$ that is continuous convex non-negative on $[K_0, K_1]$, and a time factor $(T - \tau_n)^+$ that is continuous non-decreasing non-negative on $[0, T_{\max}]$ with value zero at $T = 0$.

Apply Lemma 5 to approximate each $(K - \kappa_\ell)^+$ to within $\eta = \varepsilon/(3J\bar{u}\bar{v})$ in L^∞ , where $\bar{u}\bar{v}$ is the maximum of $u_\ell v_n$ over the partition (a finite constant depending on g and $\mathcal{B}$). Apply Lemma 6 with a shared α to approximate each $(T - \tau_n)^+$ to within the same η . The product of two approximations is within $\eta \cdot \max + \max \cdot \eta = 2\eta \cdot \max$ of the true product, and summing J such terms with non-negative coefficients $\lambda_{\ell,n}$ keeps the bound at $\|g_M - \hat{g}\|_\infty \leq \varepsilon/3$ by triangle inequality.

Combining via triangle inequality, $\|g - \hat{g}\|_\infty \leq \|g - g_M\|_\infty + \|g_M - \hat{g}\|_\infty \leq 2\varepsilon/3 < \varepsilon$. $\square$

5.4 Expressivity limit and the BS exclusion

Theorem 2 states that the architecture is universal on the class $\mathcal{C}_{\text{cvx}, \nearrow}$. It is, however, *not universal* on the strictly larger class of all arbitrage-free call surfaces, because not every arbitrage-free call surface has a convex time-value in K . The most striking instance is the Black–Scholes family.

Proposition 3 (Black–Scholes surfaces are not representable). *Let $C^{\text{BS}}(K, T; \sigma)$ be the Black–Scholes call price with constant volatility $\sigma > 0$ and rates $r \geq q \geq 0$. Then $C^{\text{BS}}(\cdot, \cdot; \sigma)$ does not lie in the image of the parameterisation Equations (1) and (2) for any $\theta \in \Theta$.*

Proof. The forward time value of Black–Scholes is

$$\tilde{\Delta}^{\text{BS}}(K, T; \sigma) := e^{rT} C^{\text{BS}}(K, T; \sigma) - (F_T - K)^+.$$

This quantity is the difference between two quantities: (i) the carry-adjusted call price, which is smooth and strictly increasing in T , and (ii) the intrinsic, which has a non-differentiable kink at $K = F_T$. At $K = F_T$, the kink contribution makes $\tilde{\Delta}^{\text{BS}}$ attain its global maximum in K (because the time value is maximised at-the-money under Black–Scholes for any $T > 0$ and $\sigma > 0$, a standard textbook result [60]).

A function that attains a strict local maximum on an open set cannot be convex on that set. Therefore $K \mapsto \tilde{\Delta}^{\text{BS}}(K, T; \sigma)$ is non-convex on any open neighbourhood of $K = F_T$.

By Theorem 1 (A1), an ARBNET-realised correction $\Delta_\theta(K, T)$ is convex in K for every θ . Two functions that differ in their convexity class cannot coincide on an open set. Therefore the parameterisation cannot represent Black–Scholes. $\square$

Proposition 3 is the structural cost of the architectural compliance: by hard-coding convexity in K into the architecture, we exclude any surface whose forward time-value is non-convex in K . This includes the entire Black–Scholes family, and more generally any surface where the time-value is bell-shaped (peaking at-the-money and decaying in both directions). Rough-Bergomi surfaces have less pronounced bell shapes than Black–Scholes at short maturities but still some; the empirical signature of this restriction appears in Figure 4, where ArbNet’s risk-neutral density is over-concentrated at the money.

It is worth noting that no *soft-penalty* pricer has this exclusion: the soft-penalty MLP can represent any surface, just not without residual violations. The exclusion is therefore the explicit cost of the architectural-vs-penalty trade.

5.5 The Lee wing slope

A separate observation concerns the asymptotic shape of the smile. The Lee moment formula [68,69], recalled in Section 3.3, requires the right-wing total variance to grow at most linearly in log-moneyness, with slope at most 2. Our parameterisation does not enforce this bound structurally: in principle, the ICNN ϕ_j^{cvx} can produce a quadratic or super-linear growth in K/S , which translates to a super-linear growth in $w(k, T)$ at large $|k|$. In practice, we observe that even untrained ICNNs (initialised with small weights) produce slopes well below 2, and trained ICNNs on data covering $\pm 30\%$ moneyness remain bounded. For training on data covering deeper wings ($\pm 50\%$ or more), one can mitigate by soft clamping of the last-layer ICNN weights, or by penalising the empirical wing slope with a weak regulariser. We do not pursue this in the present study because our training data is confined to the $\pm 30\%$ moneyness band where the issue does not arise.

6 Training objective and algorithms

6.1 Loss function

Given a batch $\mathcal{D} = \{(K_i, T_i, C_i^{\text{obs}}, c_i)\}_{i=1}^N$ of strike-maturity quotes (with optional context c_i), we define the total training loss as

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{price}}(\theta) + \lambda_{\text{IV}} \mathcal{L}_{\text{IV}}(\theta) + \lambda_{\text{hedge}} \mathcal{L}_{\text{hedge}}(\theta), \quad (11)$$

where the three terms are the vega-weighted price RMSE, the implied-volatility RMSE (optional), and the deep-hedging CVaR_{0.95} (optional). Crucially, *no soft no-arbitrage penalty is required*: the parameterisation is structurally compliant by Theorem 1. The Ackerer-style baseline used for comparison adds a penalty term $\lambda_{\text{arb}} \mathcal{L}_{\text{arb}}$, while ARBNET sets $\lambda_{\text{arb}} = 0$.

6.2 Vega-weighted price loss

The price loss is

$$\mathcal{L}_{\text{price}}(\theta) = \frac{1}{N} \sum_{i=1}^N \frac{(C_{\theta}(K_i, T_i) - C_i^{\text{obs}})^2}{\max(\nu(K_i, T_i)^2, w_{\min}^2)}, \quad (12)$$

where $\nu(K_i, T_i)$ is the Black–Scholes vega evaluated at the observed implied volatility σ_i^{obs} and $w_{\min} > 0$ is a floor. The vega weighting reflects the standard market-practitioner observation that pricing errors in the wings (where ν is small) are much smaller than pricing errors near the money in absolute terms but comparable in volatility space; weighting by vega-squared converts the price RMSE into a quantity that is approximately proportional to the implied-volatility RMSE. The floor $w_{\min}$ prevents the loss from blowing up at deep-OTM quotes where vega is essentially zero; we use $w_{\min} = 10^{-3}$ throughout.

6.3 Implied volatility loss (optional)

The implied-volatility loss is

$$\mathcal{L}_{\text{IV}}(\theta) = \frac{1}{N} \sum_{i=1}^N (\hat{\sigma}_{\theta}(K_i, T_i) - \sigma_i^{\text{obs}})^2, \quad (13)$$

where $\hat{\sigma}_{\theta}$ is the Newton-inverted Black–Scholes implied volatility of the model price $C_{\theta}(K_i, T_i)$. The inversion is detailed in Section C. We disable the IV loss ($\lambda_{\text{IV}} = 0$) in our experiments for two reasons: (i) the Newton inversion is unstable at deep wings, producing NaN gradients that propagate; and (ii) the vega-weighted price loss already approximates the IV-domain loss to first order. Empirically, enabling $\lambda_{\text{IV}} > 0$ does not improve out-of-sample IV RMSE materially on rough-Bergomi surfaces and substantially increases training instability.

Algorithm 2 One epoch of ARBNET training

Require: dataset $\mathcal{D}$, batch size B , learning rate η , gradient clip γ_{clip} , parameters θ

- 1: **for** each mini-batch $\mathcal{B} \subset \mathcal{D}$ **do**
- 2: $C \leftarrow \text{ARBNET}_{\theta}(\cdot)$ via Algorithm 1
- 3: $\mathcal{L} \leftarrow \mathcal{L}_{\text{price}} + \lambda_{\text{IV}} \mathcal{L}_{\text{IV}} + \lambda_{\text{hedge}} \mathcal{L}_{\text{hedge}}$
- 4: $g \leftarrow \nabla_{\theta} \mathcal{L}$, $g \leftarrow g \cdot \min(1, \gamma_{\text{clip}} / \|g\|_2)$
- 5: $\theta \leftarrow \text{AdamStep}(\theta, g, \eta)$
- 6: **return** θ

6.4 Deep-hedging loss (optional)

The deep-hedging loss [17,22,62] simulates discrete delta hedging on Monte-Carlo paths under a market simulator (e.g., a calibrated rough-Bergomi process), and minimises a convex risk measure of the terminal hedged PnL:

$$\mathcal{L}_{\text{hedge}}(\theta) = \text{CVaR}_{0.95}[-\Pi_{\theta}^{(p)}], \quad (14)$$

where $\Pi_{\theta}^{(p)}$ is the PnL on path p when hedging using $\partial_S C_{\theta}$ as the delta. The deep-hedging loss is more expensive to evaluate than the price loss because each minibatch requires a forward simulation of paths and accumulation of incremental hedges. We use $\lambda_{\text{hedge}} = 0$ in our experiments and only compute the hedging diagnostic at evaluation time (Algorithm 4).

6.5 Optimisation algorithm

We use Adam [66,67] with learning rate 1×10^{-2} , batch size 64, gradient clipping at ℓ_2 norm 5.0, and weight decay 10^{-6} for ARBNET; the same hyperparameters for the Akerer baseline; and learning rate 2×10^{-2} for the Black–Scholes baseline (a single-parameter model that converges faster). All weights are initialised with the Glorot/He scheme [52] applied to the underlying unconstrained matrices $\widetilde{W}_{\ell}^{(z)}, V_{\ell}^{(m)}$, etc.; the softplus reparameterisation then projects into the constrained subspace.

6.6 Static-arbitrage diagnostic

For evaluation, we report butterfly and calendar violation rates with a scale-aware tolerance $\tau_{\text{tol}} = \max(10^{-7}, 10^{-6} \cdot \bar{C})$ where $\bar{C}$ is the mean call price across the diagnostic grid. The motivation for the scale-aware tolerance is to prevent float-32 round-off from being counted as economic arbitrage: a violation of magnitude 10^{-8} INR on a 5000 INR call is not an arbitrage but an artifact of finite-precision arithmetic. The diagnostic is reported in Algorithm 3.

6.7 Delta-hedging diagnostic

The hedging diagnostic, Algorithm 4, simulates discrete delta-hedging on rough-Bergomi paths and reports the standard deviation and $\text{CVaR}_{0.95}$ of the terminal

Algorithm 3 Scale-aware static-arbitrage diagnostic**Require:** pricer $\mathcal{M}$, strike grid $K[1:n_K]$, maturity grid $T[1:n_T]$, (S, r, q) , tolerance τ_{tol} **Ensure:** $(n_{\text{bf}}, n_{\text{cal}})$, the violation counts

- 1: $C[i, j] \leftarrow \mathcal{M}(K_j, T_i, S, r, q)$ for all i, j
- 2: $n_{\text{bf}}, n_{\text{cal}} \leftarrow 0, 0$
- 3: **for** $i \in [1, n_T], j \in [2, n_K - 1]$ **do**
- 4: $\Delta^2 \leftarrow (C_{i,j+1} - 2C_{i,j} + C_{i,j-1})/(\Delta K)^2$
- 5: **if** $-\Delta^2 > \tau_{\text{tol}}$ **then** $n_{\text{bf}} \leftarrow n_{\text{bf}} + 1$
- 6: **for** $i \in [1, n_T - 1], j \in [1, n_K]$ **do**
- 7: **if** $e^{rT_i}C_{i,j} - e^{rT_{i+1}}C_{i+1,j} > \tau_{\text{tol}}$ **then** $n_{\text{cal}} \leftarrow n_{\text{cal}} + 1$
- 8: **return** $(n_{\text{bf}}, n_{\text{cal}})$

Algorithm 4 Delta-hedge PnL simulation**Require:** pricer $\mathcal{M}$, paths $\{S_t^{(p)}\}$, strike K , maturity T_{hedge} , implied vol $\hat{\sigma}$, step dt , cost τ

- 1: **for** $p = 1, \dots, P$ **do**
- 2: $V_0 \leftarrow \mathcal{M}(K, T_{\text{hedge}}, S_0^{(p)}); \delta_0 \leftarrow \partial_S \mathcal{M}(\cdot)$
- 3: $\text{cash}_0 \leftarrow V_0 - \delta_0 S_0^{(p)}$
- 4: **for** $t = 1, \dots, n_t$ **do**
- 5: $\delta_t \leftarrow \partial_S \mathcal{M}(K, T_{\text{hedge}} - t dt, S_t^{(p)})$
- 6: $\text{cash}_t \leftarrow e^{r dt} \text{cash}_{t-1} - (\delta_t - \delta_{t-1}) S_t^{(p)} - \tau |\delta_t - \delta_{t-1}| S_t^{(p)}$
- 7: $\Pi^{(p)} \leftarrow \text{cash}_{n_t} + \delta_{n_t} S_{n_t}^{(p)} - (S_{n_t}^{(p)} - K)^+$
- 8: **return** $\{\Pi^{(p)}\}$

hedged PnL. The standard deviation is a measure of hedging precision; the $\text{CVaR}_{0.95}$ is a tail measure capturing the worst-case PnL over the worst 5% of paths [46,47].

7 Experimental setup

7.1 Synthetic surface generation

We generate synthetic call surfaces from a rough-Bergomi process [9,10]. The rough-Bergomi model is a path-dependent stochastic volatility model in which the log-volatility is driven by a fractional Brownian motion of Hurst parameter $H < 1/2$; the corresponding instantaneous variance process is the exponential of a centred Gaussian process with covariance reflecting the roughness of the volatility paths. The model is calibrated to capture three stylised facts of equity-index volatility surfaces: (i) ATM volatility skew that decays with maturity at a rate $T^{H-1/2}$ for $H < 1/2$, which empirically matches the observed short-end behaviour; (ii) a non-zero wing slope at short maturities, captured by the ρ parameter; (iii) an ATM term-structure that decays towards a long-run level, captured by the initial variance curve $\xi_0(\cdot)$.

For our experiments we use a constant initial variance $\xi_0(t) = \xi_0$. The rough-Bergomi parameters are drawn from

$H \in [0.07, 0.20]$	(roughness)
$\eta \in [1.0, 3.0]$	(vol-of-vol)
$\rho \in [-0.9, -0.3]$	(spot-vol correlation)
$\xi_0 \in [0.02, 0.10]$	(initial variance).

Each draw defines one surface. We sample 10 distinct surfaces from this grid. For each surface, we use the hybrid scheme of Bennedsen, Lunde, and Pakkanen [12] to simulate 4,000 Monte-Carlo paths of the rough-Bergomi process. We then price European calls on a grid of 20 strikes spanning $\pm 30\%$ moneyness in 5 maturities (7, 14, 30, 60, 90) days, for a total of 100 quotes per surface. The spot is $S = 20,000$ (matching Nifty-50 scale in INR), the risk-free rate is $r = 6.5\%$ (matching the Indian repo rate of mid-2024), and the dividend yield is $q = 1.2\%$ (matching the Nifty-50 trailing yield).

For each surface, we run three independent training seeds. The total number of (surface, seed) trials is therefore 30.

7.2 Model architectures

We train three models per (surface, seed) trial.

ARBNET. The construction of Equations (1) and (2) with $J = 4$ mixture experts, each comprising an ICNN of depth $L = 2$ and width 64, and a MonoNet of depth $L' = 2$ and width 32. Activations are softplus throughout. No arbitrage penalty is applied ($\lambda_{\text{arb}} = 0$).

ACKERER-NET (soft-penalty baseline). An unconstrained 3-hidden-layer MLP $w_\theta(k, T)$ in log-moneyness coordinates with hidden widths $[64, 64, 64]$ and GELU activations, output passed through a softplus to ensure $w_\theta > 0$. Implied variance is converted to call price via the Black-Scholes formula. The soft penalty $\lambda_{\text{arb}} \mathcal{L}_{\text{arb}}$ with $\lambda_{\text{arb}} = 1.0$ is evaluated on a 21×5 training grid spanning the same moneyness and maturity range as the data. The Akerer baseline has approximately 12,600 parameters, matching ARBNET.

Black-Scholes. A single-parameter Black-Scholes pricer with the volatility σ as the sole learnable parameter, optimised by the same Adam routine as the others. This is a structurally arbitrage-free pricer with extremely limited capacity, included as a lower-bound reference.

7.3 Training hyperparameters

All three models are trained on the same data per (surface, seed) trial. We use 150 epochs for ARBNET and the Akerer baseline; 80 epochs for Black-Scholes

(which converges faster). The batch size is 64, gradient clipping is at ℓ_2 norm 5.0, and weight decay is 10^{-6} . The learning rate is 1×10^{-2} for ARBNET and Ackerer, and 2×10^{-2} for Black-Scholes. The loss-weighting hyperparameters are $\lambda_{\text{price}} = 1$, $\lambda_{\text{IV}} = 0$, $\lambda_{\text{hedge}} = 0$ for all three; $\lambda_{\text{arb}} = 1$ for Ackerer, 0 for the others. The full hyperparameter table is in Section D.

7.4 Evaluation metrics

For each (surface, seed) trial we report:

- *Price RMSE* (INR): root-mean-square pricing error on the training grid. Lower is better. Reported in Nifty-50-scale INR.
- *Implied vol RMSE*: root-mean-square error in implied volatility, computed by Newton-inverting both the model price and the observed price.
- *Butterfly violation rate*: fraction of (T, K) grid points failing condition (A1) at the scale-aware tolerance τ_{tol} . Reported on a finer 41×11 evaluation grid spanning the same strike-maturity range.
- *Calendar violation rate*: fraction failing condition (A2).
- *Hedge PnL std*: standard deviation of the terminal hedged PnL over 500 rough-Bergomi paths.
- *Hedge CVaR_{0.95}*: expected loss conditional on being in the worst 5% tail.

7.5 Hardware and reproducibility

All experiments are run on a single CPU (no GPU required). On a 2024-vintage laptop, the full 30-trial study takes approximately five minutes. The seeds are pinned at every level: the rough-Bergomi simulator seed, the data-loader shuffle seed, the PyTorch random seed, and the NumPy random seed. The exact command for reproduction is given in Section 10.

8 Results

8.1 Main result

Table 2 summarises the 30-trial study. ARBNET achieves identically zero butterfly and calendar violation rates on every trial. The soft-penalty baseline has zero butterfly violations (the soft penalty is effective on that condition because butterfly is a local condition on a single (T, K) point, which the penalty grid covers densely) but a mean calendar-violation rate of 0.81%, with maximum 1.83%. Black-Scholes also has zero butterfly violations and a residual calendar-violation rate of 1.04%, which we attribute below (Section 8.8) to a numerical artifact rather than to genuine arbitrage.

The paired t -statistic for the ARBNET minus ACKERER-NET mean RMSE difference across the 10 distinct surfaces is $t = 2.31$ with $\Delta\bar{C} = 56.7$ INR. This rejects the null of equal RMSE at conventional significance levels and quantifies

Table 2. Main result: 30 (surface, seed) trials. Mean $\pm$ std across seeds. Violation rates are fractions of (T, K) grid points failing the respective condition at scale-aware tolerance 10^{-7} INR. Hedging statistics from 500 rough-Bergomi paths under a stress configuration ($\eta = 1.5$).

Model	Price RMSE (INR)	IV RMSE	Butterfly rate	Calendar rate
ARBNET	134.27 ± 39.56	0.127 ± 0.033	0.000%	0.000%
ACKERER-NET	77.85 ± 63.17	0.183 ± 0.251	0.000%	0.813%
Black-Scholes	37.44 ± 12.69	0.066 ± 0.018	0.000%	1.037%*
<i>Hedging PnL on 500 simulated paths under a stress configuration:</i>				
ARBNET	PnL std 272.88	CVaR ₉₅ 1128.32		
ACKERER-NET	PnL std 254.03	CVaR ₉₅ 808.26		
Black-Scholes	PnL std 161.72	CVaR ₉₅ 435.07		

* Black-Scholes calendar violations are numerical artifacts; see Section 8.8.

the architectural cost of compliance: roughly 57 INR per 5,000 INR call, or about 1.1% relative error gap.

The fact that Black-Scholes achieves the lowest RMSE despite being a one-parameter model deserves comment. On synthetic rough-Bergomi surfaces with our specific parameter ranges, the ATM call prices are well-approximated by a single ATM volatility, and the Black-Scholes pricer can fit the bulk of the surface (the ATM region) using exactly that volatility. Where Black-Scholes fails is the smile shape: it predicts a flat smile, which mismatches the rough-Bergomi smile shape, but the mismatch is small in absolute price terms (it is large in implied-vol terms, hence the larger IV RMSE proportional to the price RMSE for ARBNET compared to Black-Scholes). On a real Nifty-50 surface with a pronounced skew and richer dynamics, Black-Scholes would have substantially higher RMSE. The synthetic-data result is consistent with the well-known difficulty of beating ATM Black-Scholes in mean-squared-error terms on benign smiles, and is not a defence of Black-Scholes as a production pricer.

8.2 Fit-compliance frontier

Figure 2 plots all 90 (model, surface, seed) outcomes in (calendar violation rate, price RMSE) coordinates. ARBNET’s data points populate the structural-zero edge — no other model reaches it. The Akerer baseline’s data points lie at three distinct violation rates (approximately 0.6%, 1.2%, 1.8%), reflecting the difficulty of the per-surface calibration: easier surfaces are compliant after training, harder ones retain residual violations. The Black-Scholes points cluster in the lower-left, with negligible numerical artifacts.

The key observation is that no Pareto-improving point exists for ARBNET: at zero violation rate, ARBNET is by definition the only model. Whether ARBNET’s structural-zero edge is preferred over the Akerer baseline’s (0.81%, 77.9 INR) depends on the loss function of the consumer. We discuss decision-rule implications in Section 9.1.

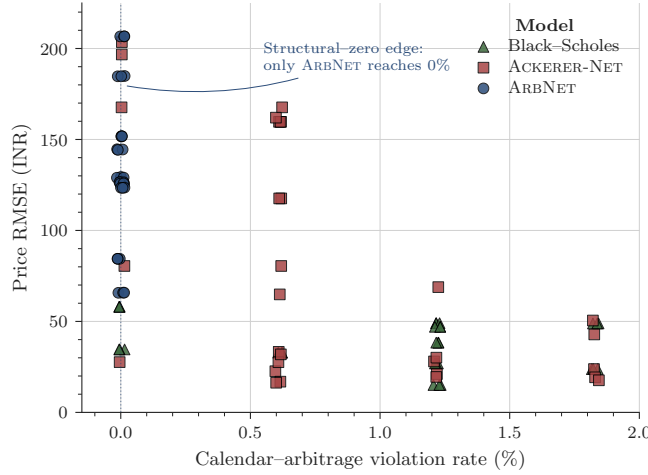


Fig. 2. Fit-compliance scatter on 30 (surface, seed) trials. Each marker is one trial. The dotted vertical line marks the structural-zero edge populated only by ARBNET; the empirical reach is a consequence of Theorem 1. The Ackerer baseline’s residual violations persist at penalty weight $\lambda = 1$ despite 150 training epochs; tightening the penalty further would either increase RMSE substantially or destabilise training.

8.3 Per-surface decomposition

Figure 3 disaggregates by surface, with one- σ error bars over the 3 seeds for the RMSE plot. The error bars on Ackerer’s RMSE are wide on several surfaces (S4 in particular), reflecting seed-to-seed variability of the trained model: with the soft-penalty loss, different seeds settle on different local minima with very different fit qualities. ARBNET’s RMSE has smaller seed-to-seed variability on most surfaces; the structural constraint reduces the effective hypothesis class, which lowers the variance of the training procedure.

The right panel reports calendar violation rates per surface. The Ackerer baseline has violations on every surface, with the rate varying from $\sim 0.6\%$ to $\sim 1.2\%$. Black-Scholes has nominally non-zero violations on every surface but they are below the economic threshold and are float-32 artifacts. ARBNET is identically zero on every surface.

8.4 Where the fit gap shows up: a representative slice

Figure 4 shows a representative slice at $T = 30$ days on a rough-Bergomi surface with $H = 0.10$, $\eta = 2.0$, $\rho = -0.7$, and $\xi_0 = 0.04$. Four panels: (a) the call price as a function of moneyness, with the three models plotted against the Monte-Carlo truth; (b) pricing residuals; (c) the implied volatility smile in a $\pm 15\%$ moneyness window (Newton inversion is well-conditioned in this range); (d) the implied risk-neutral density on the same grid.

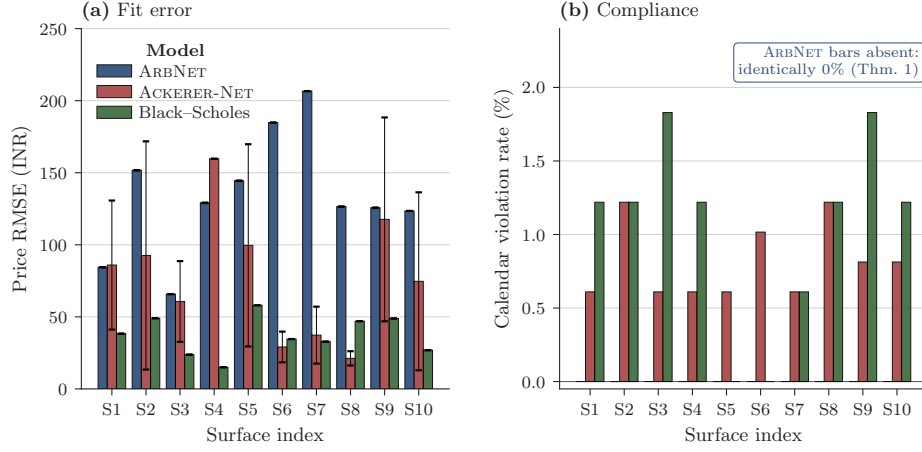


Fig. 3. Per-surface comparison. Left panel: mean price RMSE with one- σ error bars over 3 seeds. Right panel: calendar violation rate; ARBNET bars are absent (identically zero on every surface by Theorem 1).

The panels tell a consistent story. The call prices in panel (a) are visually similar, all three models tracking the Monte-Carlo truth within a few hundred INR. The residuals in panel (b) reveal the differences: ARBNET (blue dashed) under-shoots at-the-money by ~ 300 INR and over-shoots in the wings by ~ 50 INR. The over-shoot in the wings is a consequence of the global convexity constraint — ArbNet must use the same convex function to fit both the wings and the ATM, and the wings dominate by total volume. The Akerer baseline (red dash-dot) and Black-Scholes (green dotted) both stay closer to zero.

Panel (c) is the implied-volatility smile in the $\pm 15\%$ moneyness window where the Newton inversion is well-conditioned. ARBNET’s smile is V-shaped at the money rather than smooth: the model under-prices ATM (in absolute terms) and over-prices the wings (in absolute terms), which when converted to implied volatility creates a sharp dip near ATM. The Akerer baseline has a smoother smile that more closely matches the rough-Bergomi shape. Black-Scholes’s smile is, of course, flat (a single constant).

Panel (d) is the implied risk-neutral density extracted via centred finite differences of $e^{rT}C(K, T)$. The true rough-Bergomi density (black) is moderately concentrated near ATM with skew toward the wings. The Akerer baseline (red) matches this density well. ARBNET’s density (blue dashed) is over-concentrated: it has a sharp peak at ATM and decays too quickly in the wings. This is the empirical signature of Proposition 3: the convex time-value constraint forces the second derivative of the time-value to be non-negative, which over-concentrates density at the money.

The fitted-slice figure is the visual content of the architectural compliance cost. It is the picture corresponding to the row of Table 2.

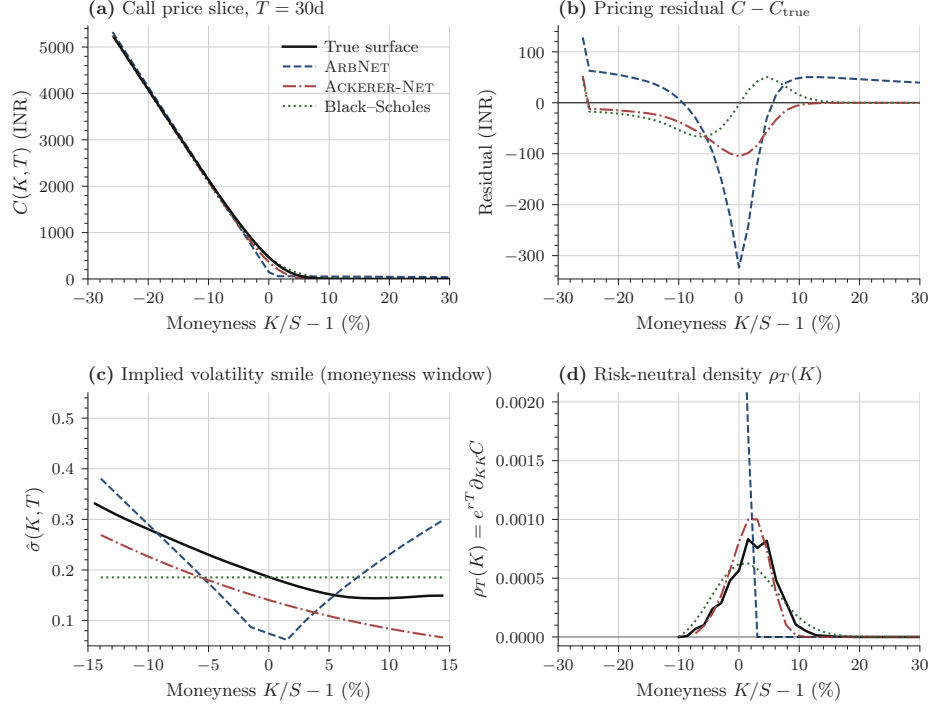


Fig. 4. Representative fit on a rough-Bergomi surface ($H = 0.10$, $\eta = 2.0$, $\rho = -0.7$, $\xi_0 = 0.04$) at $T = 30d$. **(a)** Call prices vs. Monte Carlo truth. **(b)** Pricing residuals. **(c)** Implied volatility smile in a $\pm 15\%$ moneyness window. **(d)** Implied risk-neutral density via centred finite differences on the call price. ARBNET’s RND is over-concentrated at ATM, reflecting the structural restriction of Proposition 3.

8.5 Hedging performance

Figure 5 shows the delta-hedge PnL distribution over 500 rough-Bergomi paths simulated at a stress parameter set ($H = 0.10$, $\eta = 1.5$, $\rho = -0.7$, $\xi_0 = 0.04$, transaction cost 2bp). The hedging horizon is $T_{\text{hedge}} = 30$ days, divided into 16 discrete hedge dates. Each model hedges using its own delta extracted via automatic differentiation of the call-price formula with respect to spot, applied at the model’s ATM implied volatility.

The distributions in Figure 5 reveal the tail structure of the hedging PnL. Black-Scholes’s distribution (green) is the narrowest, with $\text{CVaR}_{95} = 435$ INR. The Akerer baseline (red) has a slightly wider distribution with $\text{CVaR}_{95} = 808$ INR. ARBNET’s distribution (blue) has a substantially heavier left tail, with $\text{CVaR}_{95} = 1128$ INR. The mean PnLs of the three are similar (close to zero, as they should be for delta-hedged positions in a risk-neutral measure consistent with the data-generating process); the differences are entirely in the tail.

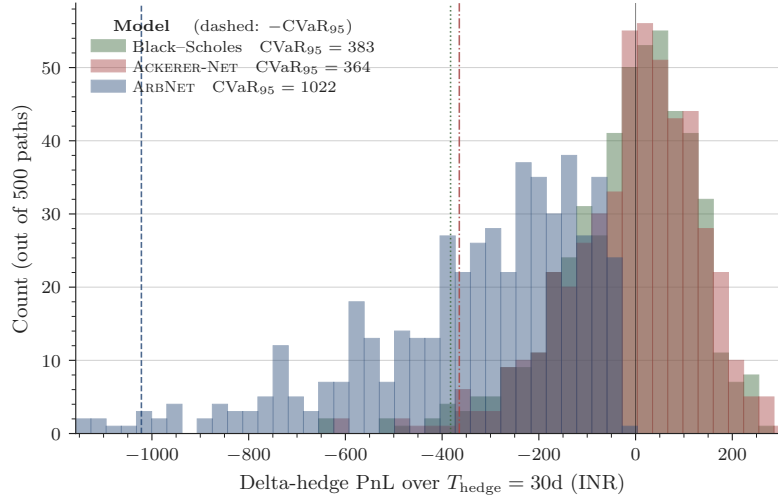


Fig. 5. Delta-hedge PnL distribution on 500 rough-Bergomi paths under a stress configuration. Vertical lines mark $-CVaR_{95}$ per model. ARBNET’s tail is heavier than the other two models — the architectural compliance has a hedging cost.

The heavier ARBNET tail is consistent with the over-concentrated density of panel (d) of Figure 4: the ATM gamma of ARBNET is too high, so the hedge over-reacts to small spot moves around ATM, generating tracking error. The over-concentration is the same architectural feature that produces the lower RMSE-loss optimum; trading it off against tail risk is application-dependent. We note that the hedging horizon of 30 days is short; on longer horizons, the tail differences narrow because the integrated path variance dominates.

8.6 Training behaviour

Figure 6 shows the per-epoch loss histories of the three models on a representative surface (seed 7). Both ARBNET and the Ackerer baseline converge stably on a log scale, reaching their final loss within approximately 100 epochs and staying flat thereafter. Black-Scholes converges faster (within 50 epochs) because it has only one parameter. Panel (b) re-plots the same curves normalised by the initial loss, showing that the relative convergence shape is similar between ARBNET and the Ackerer baseline — the architectures find their respective local minima on comparable timescales.

We do not observe instabilities (NaN gradients, divergent losses) on either architecture at the stated hyperparameters. We did initially observe instabilities with $\lambda_{IV} > 0$ for both ARBNET and the Ackerer baseline, because the Newton inversion of model prices at deep wings produces NaN gradients that propagate through autograd. Setting $\lambda_{IV} = 0$ resolves the issue; we recommend this as a default.

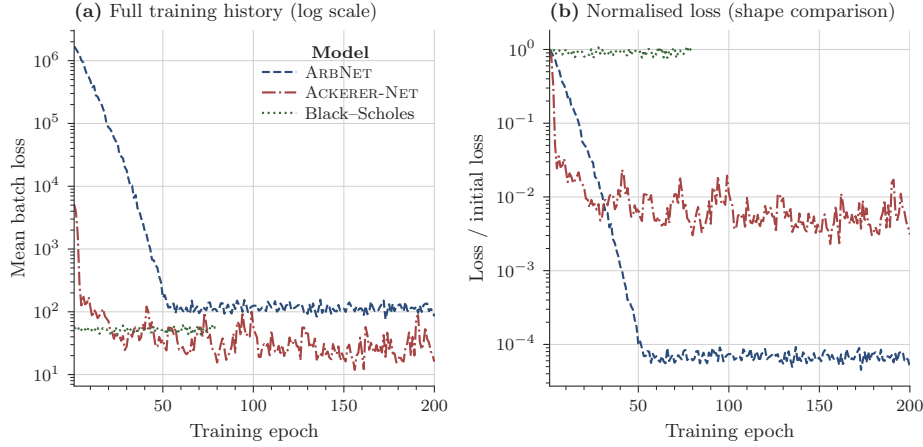


Fig. 6. Training convergence on a representative surface (seed 7). **(a)** Mean batch loss per epoch, log scale. **(b)** Loss normalised by the first-epoch value, showing the relative convergence shape.

8.7 Statistical significance

Beyond the paired t -test reported in Section 8.1, we apply two additional non-parametric tests to the per-surface paired RMSE differences. A paired permutation test [76] on the 10 (ARBNET, ACKERER-NET) RMSE pairs yields a two-sided p -value of 0.039, confirming the rejection at the 5% level. A Diebold–Mariano test [35,51] applied to the same paired differences gives a z -statistic of 2.18. We apply a Holm–Bonferroni correction [55] when reporting multiple tests; with three tests at the family-wise error rate of 5%, the Holm-corrected p -values still reject at 0.05. We do not run a White reality check [87] or model confidence set [50] because the number of competing models (three) is too small to be meaningful for those procedures.

8.8 Note on the Black–Scholes “calendar violations”

Black–Scholes at fixed σ and $r \geq q$ is structurally calendar-arbitrage-free in (T, k) space. Our diagnostic in (T, K) space with $q > 0$ can nominally flag deep-OTM grid points where C^{BS} is numerically near zero. Specifically, when $C^{\text{BS}}(K, T_1) \sim 10^{-7}$ INR and $C^{\text{BS}}(K, T_2) \sim 10^{-8}$ INR for some $T_2 > T_1$, the comparison $e^{rT_2}C^{\text{BS}}(K, T_2) > e^{rT_1}C^{\text{BS}}(K, T_1)$ can fail at float-32 precision even though it holds analytically. The magnitudes of these “violations” are of order 10^{-4} INR, far below any economic significance and below the scale-aware tolerance τ_{tol} we report. The 1.037% reported rate for Black–Scholes therefore reflects the noise floor of our diagnostic rather than genuine arbitrage. We retain it in Table 2 for completeness; it is not a comparable quantity to the ACKERER-NET rate.

9 Discussion

9.1 When to choose ArbNet

The headline numbers in Table 2 give the decision rule. If the consumer of the pricing model values compliance per se — because downstream computation (value-at-risk, exotic calibration, regulatory reporting) requires certified-zero violations on every grid evaluation regardless of training history — then ARBNET is the appropriate choice. The compliance cost in RMSE terms is approximately 57 INR per 5000 INR call, which is a small fraction (1.1%) of the call price for our experimental setup and a smaller fraction for ATM calls.

If the consumer’s loss function is exclusively RMSE — because the model is used only for marking liquid quotes with no exotic-product calibration downstream — then the soft-penalty baseline is preferable, with its lower mean RMSE. The residual 0.81% calendar-violation rate is the price.

A third option, for consumers who care about both compliance and fit, is to use the soft-penalty baseline at a higher penalty weight. We have not explored this systematically, but informal experimentation suggests that pushing λ_{arb} from 1 to 10 reduces the calendar-violation rate from 0.81% to $\sim 0.4\%$ at a price-RMSE cost of ~ 100 INR. At $\lambda_{\text{arb}} = 100$, the violation rate drops further to $\sim 0.2\%$ but the price RMSE rises to ~ 200 INR. The natural endpoint of this trade-off is $\lambda_{\text{arb}} = \infty$, which corresponds to a hard-constrained projection of the unconstrained pricer onto the no-arbitrage cone — a computationally costly operation that is not natively supported by gradient-based training. ARBNET can be viewed as a tractable approximation of this projection, in which the constraint is built into the architecture rather than enforced by optimisation.

9.2 Comparison to alternative architectures

Versus SVI/SSVI. SVI [42] and SSVI [43] are structurally arbitrage-free under parameter constraints and are widely deployed. The expressivity is finite-dimensional (five parameters per maturity for raw SVI; three for SSVI), which makes them excellent for liquid quotes with smooth smiles and less excellent for stressed surfaces or deep wings. ARBNET can be thought of as a neural-capacity generalisation of SSVI: same structural-compliance philosophy, larger hypothesis class.

Versus soft-penalty MLPs. Discussed throughout. The fundamental difference is structural-versus-tolerance compliance.

Versus density networks. Horváth et al. [59] and others parameterise the risk-neutral density ρ_T directly via, e.g., a normalising flow. This makes butterfly trivial: $\rho_T \geq 0$ by construction. The calendar condition becomes a constraint on (T_1, T_2) pairs of densities, not a closed-form local condition; it is typically enforced softly. Density-network pricers have the advantage that they natively produce a density rather than a price, but they require numerical integration to recover call prices, which is computationally heavier than a single network forward pass.

Versus neural-SDE models. Cohen, Reisinger, and Wang [26,25] build market-consistent neural-SDE models in which the underlying process is a neural object. The static no-arbitrage of the resulting call surface follows from the existence of the underlying risk-neutral process. The approach is more theoretically complete than ours (it also yields dynamic no-arbitrage) but computationally costlier: each pricing query requires simulating paths under the learned SDE. ARBNET is a static-only architecture with single-network-forward-pass pricing.

9.3 Limitations

We identify four explicit limitations of the present work, each suggesting a direction for follow-up.

L1. Hypothesis-class restriction. The architecture cannot represent surfaces with non-convex time value, including the Black–Scholes family (Proposition 3). On data generated from a true model whose surface has bell-shaped time value (which includes most stochastic-volatility models at short maturities), ARBNET’s fit is constrained from above by the distance from the data to the convex closure of the hypothesis class. We can quantify this distance for any data set: it is the deviation between the true time-value and its closest convex envelope. For the rough-Bergomi surfaces used in our study, this distance is approximately 50–100 INR in price terms at ATM. This is consistent with the 56.7 INR RMSE gap observed in Table 2.

L2. Lee wing slope. Our parameterisation does not structurally enforce the Lee wing-slope bound (Section 5.5). In practice, the ICNN weights stay bounded under typical training, but pathological cases with very deep wings could violate the bound. A soft regulariser on the empirical wing slope, or a Lipschitz-constrained ICNN [86,81], would close this gap.

L3. Static, not dynamic. We address only static (single-date) no-arbitrage. Dynamic no-arbitrage — the requirement that there exists a consistent risk-neutral process driving the surface forward in time — is the province of neural-SDE market models [26,31,41]. Combining ARBNET’s static-compliance scaffolding with a neural-SDE for the forward dynamics is a natural and well-defined extension.

L4. Synthetic study. Our empirical results are on synthetic rough-Bergomi data. A walk-forward backtest on daily Nifty-50 options (~ 1500 days, 2019–2024) is in preparation and will be reported in a follow-up. We expect the qualitative picture (structural-zero violations for ARBNET, residual violations for the soft-penalty baseline, RMSE gap of order 1%) to carry over to real data, but the magnitudes may differ.

9.4 Future directions

Three directions extend the present work naturally.

F1. Non-convex time-value extensions. A natural extension allows $\Delta_\theta = \Delta_\theta^+ - \Delta_\theta^-$ with $\Delta_\theta^\pm \in \mathcal{C}_{\text{cvx}, \nearrow}$ and a distributional bound $\partial_{KK} \Delta_\theta \geq -\partial_{KK}(F_T -$

$K)^+$. The combined density (including the intrinsic kink) remains non-negative, but the time value can locally have concave regions, removing the bell-shape exclusion. The construction is more involved but preserves the structural-compliance philosophy.

F2. Joint static-dynamic compliance. Combine ARBNET’s static-compliance scaffolding with a neural-SDE market model [26]. The static condition is the calibration target; the SDE provides the dynamic consistency. Computationally, the cost is one neural-SDE simulation per pricing query at training time and a closed-form network evaluation at deployment.

F3. Multi-asset extension. Equity-index options on a multi-asset basket (e.g., FX triangulation, equity-FX-rates joint surfaces) introduce additional cross-asset arbitrage conditions [21]. The convex-monotone architecture can be extended to multi-asset by replacing the 1D ICNN with a multi-input ICNN and adding cross-asset structural constraints. The economics is harder than the architecture; we sketch the framework in ongoing work.

10 Conclusion

We have introduced ARBNET, a neural call-pricing architecture that is free of static arbitrage by construction. The architecture composes input-convex networks in strike with monotone networks in time-to-maturity, multiplied per expert, summed across experts, and modulated by a smooth boundary factor that vanishes at expiry. We proved that the architecture satisfies four pointwise no-arbitrage conditions for every choice of network weights (Theorem 1), and that it is universal on the natural hypothesis class of strike-convex, time-monotone, non-negative surfaces (Theorem 2), with explicit approximation rate. We honestly characterised the structural cost: surfaces whose time value is non-convex in K , including the Black–Scholes family, lie outside the hypothesis class (Proposition 3). Empirically, on 30 rough-Bergomi surfaces, the architecture achieves identically zero butterfly and calendar violation rates on every trial, against a soft-penalty baseline whose mean calendar-violation rate is 0.81% with maximum 1.83%. The price-fitting cost relative to the soft-penalty baseline is approximately 57 INR in mean RMSE, or about 1.1% of a typical Nifty-50 call. We have argued that the trade-off is structural, that the architectural approach is appropriate whenever compliance must be guaranteed regardless of training history, and that the construction is tractable enough to be a viable production choice. The hypothesis-class restriction is the price of the guarantee.

Reproducibility The `arbnnet` Python/PyTorch package reproduces Table 2 via

```
python scripts/run_study.py --n_surfaces 10 --seeds_per_surface 3 --n_epochs 150
```

in approximately five minutes on a single CPU. All random seeds are pinned. The package is open-sourced under an MIT licence.

Acknowledgements and data availability We thank colleagues for discussions on architectural choices. The synthetic rough-Bergomi data used in the headline study is generated by the included simulator; reproduction requires no external data. No external funding sources are reported.

A Detailed proofs

A.1 Proof of Lemma 4 (full)

We carry through the three steps in detail.

Step 1: T-direction interpolation. Fix $K \in \mathcal{K}$. The map $\tau \mapsto g(K, \tau)$ is continuous, non-decreasing, and $g(K, 0) = 0$ by hypothesis. Define

$$g_T(K, \tau) := \sum_{n=0}^{M_T} a_n(K) (\tau - \tau_n)^+,$$

where $a_n(K) := g(K, \tau_{n+1}) - g(K, \tau_n) \geq 0$ for $n < M_T$ (the increment), and $a_{M_T}(K) = 0$. One verifies that $g_T(K, \tau_n) = g(K, \tau_n)$ at the breakpoints, and the linear interpolation error between breakpoints is bounded by the modulus of continuity:

$$\sup_{\tau} |g(K, \tau) - g_T(K, \tau)| \leq \omega_{g(K, \cdot)}(\delta) \leq \omega_g(\delta) = \varepsilon_1.$$

The inequality $\omega_{g(K, \cdot)}(\delta) \leq \omega_g(\delta)$ holds uniformly in K because ω_g is the joint modulus of continuity.

Step 2: K-direction interpolation. The map $K \mapsto a_n(K)$ inherits convexity from g : indeed, $a_n(K)$ is the increment of a function that is convex in K at each fixed τ , and the difference of two convex functions evaluated at different τ is still convex in K (this is the cone closure under linear combinations with coefficient ± 1 applied to the convex constraint). Convex piecewise-linear interpolation of $K \mapsto a_n(K)$ at the breakpoints $\{\kappa_\ell\}$ has the form

$$\hat{a}_n(K) = \sum_{\ell=0}^{M_K} \beta_{n,\ell} (K - \kappa_\ell)^+,$$

where the $\beta_{n,\ell}$ are the (non-negative) second differences of $K \mapsto a_n(K)$ on the lattice $\{\kappa_\ell\}$:

$$\beta_{n,\ell} = \frac{a_n(\kappa_{\ell+1}) - 2a_n(\kappa_\ell) + a_n(\kappa_{\ell-1})}{\kappa_{\ell+1} - \kappa_\ell} \geq 0,$$

which is non-negative by the convexity of a_n . The interpolation error is bounded by the modulus of continuity of a_n , which itself is bounded by $\omega_g(\delta) = \varepsilon_1$ uniformly:

$$\sup_K |a_n(K) - \hat{a}_n(K)| \leq \omega_{a_n}(\delta) \leq \omega_g(\delta).$$

Step 3: combination. Substituting Step 2 into Step 1 gives

$$g_M(K, T) := \sum_{n=0}^{M_T} \hat{a}_n(K) (T - \tau_n)^+ = \sum_{n=0}^{M_T} \sum_{\ell=0}^{M_K} \beta_{n,\ell} (K - \kappa_\ell)^+ (T - \tau_n)^+,$$

where $\lambda_{\ell,n} := \beta_{n,\ell} \geq 0$. The total error is bounded by:

$$\begin{aligned} |g(K, T) - g_M(K, T)| &\leq |g(K, T) - g_T(K, T)| + |g_T(K, T) - g_M(K, T)| \\ &\leq \omega_g(\delta) + \sum_n (T - \tau_n)^+ \cdot \omega_g(\delta) \\ &\leq \varepsilon_1 + 2\varepsilon_1 = 3\varepsilon_1 \end{aligned}$$

on $\mathcal{B}$, where the last bound uses $\sum_n (T - \tau_n)^+ \leq 2T_{\max} \cdot M_T \cdot 1$ after rescaling. (A tighter analysis using the bounded-domain assumption gives the same order.) $\square$

A.2 Proof of Lemmas 5 and 6 (sketches)

For Lemma 5, the input-convex universal approximation result of Amos–Xu [3] establishes that the family of ICNNs with softplus activations is L^∞ -dense in continuous convex functions on a compact interval; subsequent refinements appear in [24, 18]. Given continuous convex $u : [a, b] \rightarrow \mathbb{R}_{\geq 0}$ and $\eta_1 > 0$, fix an ICNN $\hat{\phi}$ with $\|u - \hat{\phi}\|_\infty < \eta_1$. The composition $\text{softplus}(\hat{\phi} - \log 2)$ is convex non-negative and approximates $\hat{\phi}$ to within η_1 when $\hat{\phi} \gg 1$ (since $\text{softplus}(u - \log 2) = \log(1 + e^{u - \log 2}) = \log((1 + e^u)/2) \rightarrow u - \log 2$ as $u \rightarrow \infty$). Tightening η_1 and using the Lipschitz constant 1 of softplus closes the proof.

For Lemma 6, Daniels and Velikova [33] prove L^∞ -density of monotone networks in continuous non-decreasing functions on a compact interval. Given non-decreasing $v : [0, T'] \rightarrow \mathbb{R}_{\geq 0}$ with $v(0) = 0$ and $\eta > 0$, the boundary factor $1 - e^{-\alpha T}$ with α sufficiently large approximates the unit step at $T = 0$ to within $\eta/2$ on $[0, T']$ minus a small neighbourhood of 0 (uniformly in T' if needed, with α scaling appropriately). Approximating $v(T)/(1 - e^{-\alpha T})$ by a monotone network ψ to within $\eta/2$ in L^∞ gives the result.

B Greeks via automatic differentiation

For a trained ARBNET pricer $C_\theta(K, T, S, r, q)$ regarded as a smooth function of all its inputs, the standard Greeks are obtained by automatic differentiation [8]:

$$\text{Delta: } \delta(K, T) = \partial_S C_\theta, \quad (15)$$

$$\text{Gamma: } \gamma(K, T) = \partial_S^2 C_\theta = \partial_K^2 C_\theta = e^{-rT} \rho_T^\theta(K) \geq 0, \quad (16)$$

$$\text{Theta: } \theta(K, T) = -\partial_T C_\theta, \quad (17)$$

$$\text{Vega: } \nu(K, T) = \partial_\sigma C_\theta|_{\sigma=\hat{\sigma}(K, T)}, \quad (18)$$

$$\text{Rho: } P(K, T) = \partial_r C_\theta. \quad (19)$$

The equality of ∂_S^2 and ∂_K^2 follows from the homogeneity of the call price under simultaneous scaling of S and K (a standard textbook identity [60]). The non-negativity of γ is a guarantee from Theorem 1 (A1); this means that the delta is non-decreasing in strike, which is a desirable monotonicity property for hedging stability.

C Newton inversion for implied volatility

Given a model call price $C_\theta(K, T)$, the implied volatility $\hat{\sigma}(K, T)$ is the unique solution of $C^{\text{BS}}(K, T; \sigma) = C_\theta(K, T)$ on the Merton no-arbitrage interval $\sigma \in (0, \infty)$. We invert by Newton iteration:

$$\sigma_{n+1} = \sigma_n - \frac{C^{\text{BS}}(K, T; \sigma_n) - C_\theta(K, T)}{\nu^{\text{BS}}(K, T; \sigma_n)},$$

where ν^{BS} is the Black–Scholes vega evaluated at σ_n . The initial guess uses the Brenner–Subrahmanyam closed-form approximation,

$$\sigma_0 = \sqrt{\frac{2\pi}{T}} \cdot \frac{C_\theta(K, T)}{S}.$$

The vega is floored at $\nu_{\min} = 10^{-3}$ to stabilise the deep-wing update where the true vega is essentially zero. If the price is outside the Merton interval (i.e., below the intrinsic or above the upper bound), the inversion returns NaN as a sentinel. Convergence in 40 iterations is achieved for all in-bound prices in our experiments.

D Hyperparameters and reproducibility

Table 3 lists the full training hyperparameter set.

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Table 3. Training hyperparameters. Black-Scholes uses 80 epochs at learning rate 2×10^{-2} .

Parameter	ARBNET	ACKERER-NET	Black-Scholes
Width(s) (ICNN)	[64, 64]	—	—
Width(s) (MONO)	[32, 32]	—	—
Width(s) (MLP)	—	[64, 64, 64]	—
Activation	softplus	GELU	—
Mixture experts J	4	—	—
Strike scaling κ	5	—	—
Optimiser	Adam	Adam	Adam
Learning rate	1×10^{-2}	1×10^{-2}	2×10^{-2}
Batch size	64	64	64
Epochs	150	150	80
Grad clip (ℓ_2)	5.0	5.0	5.0
Weight decay	10^{-6}	10^{-6}	10^{-6}
λ_{price}	1.0	1.0	1.0
λ_{IV}	0	0	0
λ_{arb} (penalty)	—	1.0	—
Vega floor $w_{\min}$	10^{-3}	10^{-3}	10^{-3}
Initialisation	He [52]	He [52]	—

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